

PTO Time Management High Level Design Document

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Application Development
October 11, 2019 - July 3, 2020

Introduction

The purpose of this High-Level Design (HLD) Document is to add the necessary detail to the current project description to represent a suitable model for coding. This document is also intended to help detect contradictions prior to coding and can be used as a reference manual for how the modules interact at a high level.

Purpose

Currently, work hours can be lost to PTO because time-off availability is manually handled and approved by managers. This project is to create a tool that allows managers to set parameters to automatically manage/control/approve how many team members in the Nashville and Schaumburg office can request PTO on a daily basis, on holidays, on a first-come first-serve basis. This tool automates time off and shift swap requests allowing the Nashville and Schaumberg management teams to effectively manage these requests. The current process for managing these requests is done through inefficient system utilizing Outlook and Outlook Calendar to manage the process. This application will replace the current system.

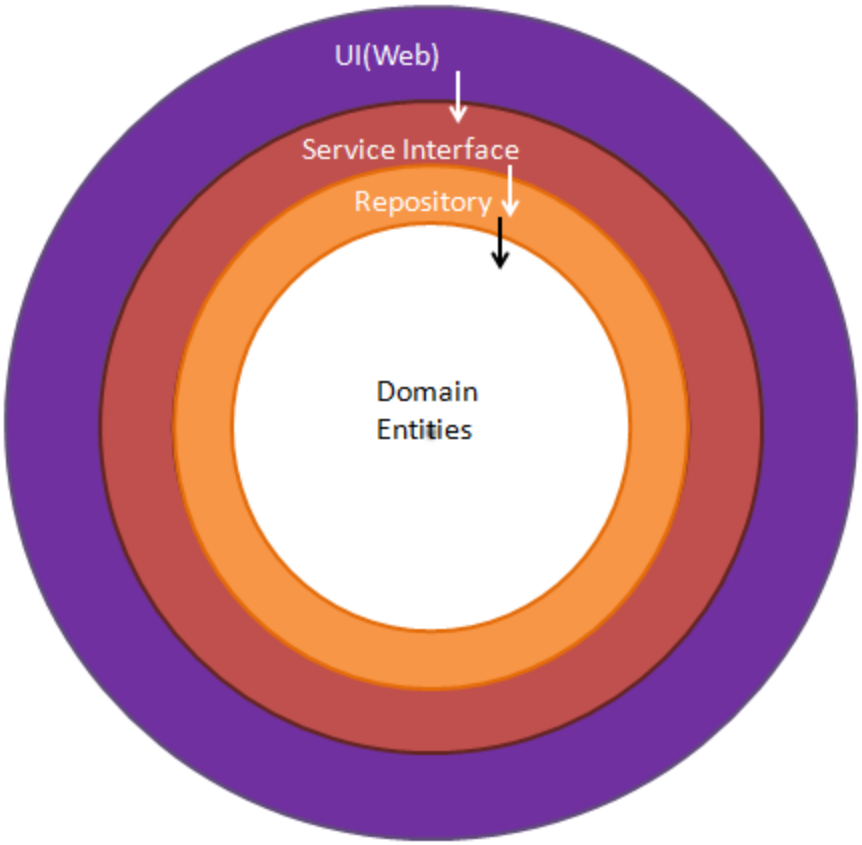
General Description

High Level Design

This application will use .NET Core 2.1 with and ReactJS front end to handle the view of the user, and an MVC entry point to manage authentication and authorization and HTTP calls from the React JS front end. The middle tier of the application will utilize the Onion Architecture Pattern to manage and separate concerns throughout the application.

Onion Architecture

Onion Architecture is an application architecture that relies heavily on the Dependency Inversion Principle. The Onion Architecture term was coined by Jeffrey Palermo in 2008. This architecture provides a better way to build applications for better testability, maintainability, and dependability on the infrastructures like databases and services. Between the layers of the Onion, there is a strong *dependency rule*: **outer layers can depend on lower layers, but no code in the lower layer can depend directly on any code in the outer layer**.

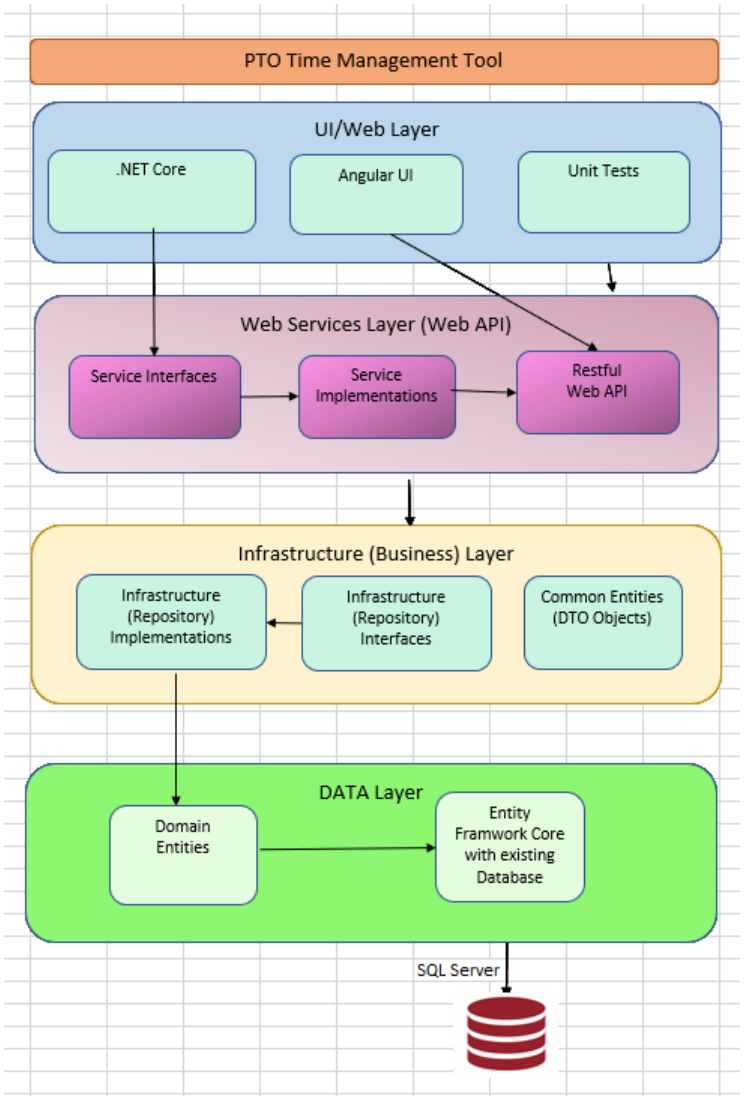


It consists of layers as :

1. Presentation layer
2. Service layer (Web API)
3. Business layer
4. Domain Entities (Data) layer.

Each layer communicates with the layer below it through interfaces. The internal layers never depend on external layer.

Architecture Diagram for Time Off Request Tool :



Projects in Application

Folder	Holds	Project Name	Description
Test	MSTest Project	PTOTMT.Test.MSTest	A .Net Core MSTest Unit Tests project that contains unit tests.
Web	Web .Net Core UI Main Project with Angular as ClientApp	PTOTMT.Web	An ASP.Net Core 3.0Test Web Application project with API Controller for a RESTful HTTP Service. This project has integrated Angular 8 application set up to create Angular 8 UI Framework along with jQuery, Bootstrap and with additional technologies to better the user experience. It is the most external part of an application by which the end user can interact with the application (User Interface). It builds loosely coupled applications with in-built dependency injection in ASP.NET Core. It represents the UI/Web layer of the Onion Architecture.
Common	Common methods in infrastructure (Business Layer) - (Data Transfer Object)	PTOTMT.Common	A .Net Core Class library project to hold Base Data Transfer Classes and methods in Business Layer. It also holds DbContext class. It represents the Business Layer of the Onion Architecture.
Repository	Repository Interfaces and implementations	PTOTMT. Repository	A .Net Core Class library project to hold interface **definitions and implementations** for data **repositories** in infrastructure

			(Business Layer) to retrieve data and manipulate data in SQL server data storage using Entity Framework. Through these objects the Business Layer interacts (passes data to) with the beneath Web Services Layer. It represents the Business Layer of the Onion Architecture.
Service	Web API	PTOTMT. Service	A .Net Core Web API project that hosts Web API End Points for all CRUD operations on all table data. It represents the Service Layer of Onion Architecture .
Service	Service Interfaces and Implementations	PTOTMT. Service. WebCalls	A .Net Core Class Library project that holds **implementations for all interfaces** created in PTOMT.Services project for the purpose of generating business objects to execute all web service call methods which would fetch secured data from already hosted web services by making http call. It represents the Service Layer of Onion Architecture .
Domain	Domain Entities to use in Data Layer	PTOTMT.Domain	A .Net Core Class library project that contains the classes for Entity Framework Code First Migration objects. It holds DbContext class. The Entity Framework Code First data access approach needs to create a data access context class that inherits from the DbContext class.

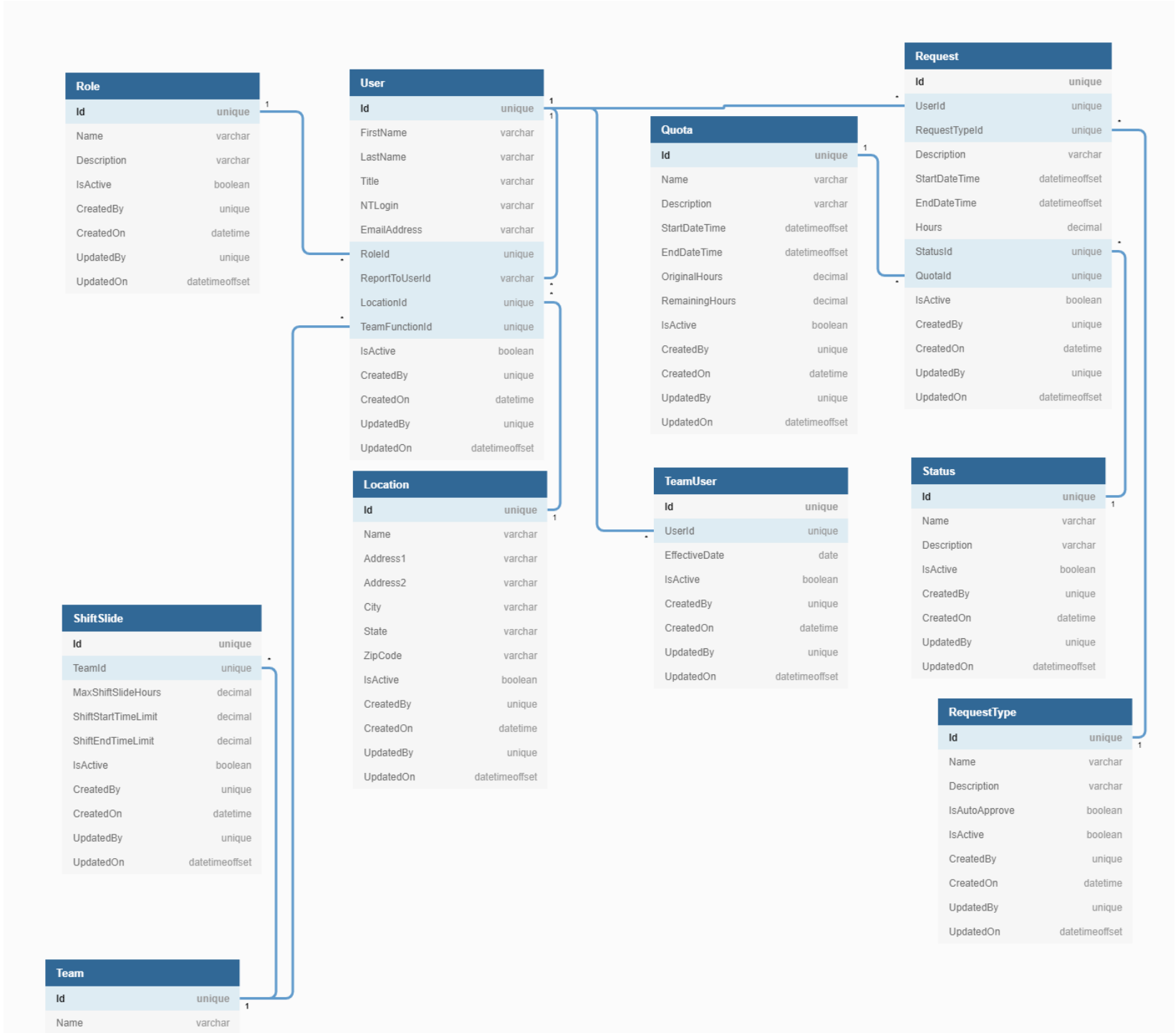
Database	Sql Server Database Project	PTOMT.Database	A Visual Studio's Sql Server Database Project to manage our database scripts. We can create a new database project and import database schema from an existing database, a .sql script file. We can then invoke the same visual designer tools (Transact-SQL Editor, Table Designer) available for connected database development to make changes to the offline database project, and publish the changes back to the production database. The changes can also be saved as a script to be published later. Using the Project Properties panel, we can change the target platform to different versions of SQL Server (including SQL Azure).

Database Schema

3 Schema

PTO	**Security**	**Config**
Request	Users	RequestType
Quota	Role	Location
Flex	Team	FlexType
		Status

1.


Role
Id unique
Name varchar
Description varchar
IsActive boolean
CreatedBy unique
CreatedOn datetime
UpdatedBy unique
UpdatedOn datetimeoffset
User
Id unique
FirstName varchar
LastName varchar
Title varchar
NTLogin varchar
EmailAddress varchar
RoleId unique
ReportToUserId varchar
LocationId unique
TeamFunctionId unique
IsActive boolean
CreatedBy unique
CreatedOn datetime
UpdatedBy unique
UpdatedOn datetimeoffset
Request
Id unique
UserId unique
RequestTypeId unique
Description varchar
StartDateTime datetimeoffset
EndDateTime datetimeoffset
Hours decimal
StatusId unique
QuotaId unique
IsActive boolean
CreatedBy unique
CreatedOn datetime
UpdatedBy unique
UpdatedOn datetimeoffset
Quota
Id unique
Name varchar
Description varchar
StartDateTime datetimeoffset
EndDateTime datetimeoffset
OriginalHours decimal
RemainingHours decimal
IsActive boolean
CreatedBy unique
CreatedOn datetime
UpdatedBy unique
UpdatedOn datetimeoffset
Location
Id unique
Name varchar
Address1 varchar
Address2 varchar
City varchar
State varchar
ZipCode varchar
IsActive boolean
CreatedBy unique
CreatedOn datetime
UpdatedBy unique
UpdatedOn datetimeoffset
TeamUser
Id unique
UserId unique
EffectiveDate date
IsActive boolean
CreatedBy unique
CreatedOn datetime
UpdatedBy unique
UpdatedOn datetimeoffset
Status
Id unique
Name varchar
Description varchar
IsActive boolean
CreatedBy unique
CreatedOn datetime
UpdatedBy unique
UpdatedOn datetimeoffset
ShiftSlide
Id unique
TeamId unique
MaxShiftSlideHours decimal
ShiftStartTimeLimit decimal
ShiftEndTimeLimit decimal
IsActive boolean
CreatedBy unique
CreatedOn datetime
UpdatedBy unique
UpdatedOn datetimeoffset
RequestType
Id unique
Name varchar
Description varchar
IsAutoApprove boolean
IsActive boolean
CreatedBy unique
CreatedOn datetime
UpdatedBy unique
UpdatedOn datetimeoffset
Team
Id unique
Name varchar

Databases

PTOTimeManagement, PTOTimeManagementTest for development and testing..

Connection String

Data Source=DESKTOP-GR4UU4J;Integrated Security=True;Persist Security Info=False;Pooling=False;MultipleActiveResultSets=False;Connect Timeout=60;Encrypt=False;TrustServerCertificate=False

Table Definitions

- A table that contains all PTO requests submitted by Sales Operations Team
- Request Status (Accepted, Cancelled, WaitList, Declined)

PTO. Request

Column Name	Data Type	Required?	Description
Id	uniqueidentifier	yes	Primary Key
UserId	uniqueidentifier	yes	Foreign Key
RequestTypeId	uniqueidentifier	yes	Foreign Key
Description	nvarchar(50)	no	
StartDateTime	datetimeoffset	yes	
EndDateTime	datetimeoffset	yes	
Hours	decimal (4,1)	yes	
StatusId	uniqueidentifier	yes	Foreign Key
QuotaId	uniqueidentifier	yes	Foreign Key
IsActive	bit	yes	
CreatedBy	uniqueidentifier	yes	Foreign Key

CreatedOn	datetimeoffset	yes	Default value as GETDATE()
UpdatedBy	uniqueidentifier	no	userId
UpdatedOn	datetimeoffset	no	Default value as GETDATE()

- A table that contains all Flex requests submitted by Sales Team
- Request Status (Accepted, Cancelled, WaitList, Pending, Declined)

PTO. Flex

Column Name	Data Type	Required?	Description
Id	uniqueidentifier	yes	Primary Key
UserId	uniqueidentifier	yes	Foreign Key
FlexTypeId	uniqueidentifier	yes	Foreign Key
Name	Nvarchar(30)	no	
Description	nvarchar(50)	no	
StartDateTime	datetimeoffset	yes	
EndDateTime	datetimeoffset	yes	
Hours	decimal (4,1)	yes	
CoWorkerId	uniqueidentifier	yes	Foreign Key
CoWorkerFlexId	uniqueidentifier	yes	Foreign Key
AnotherStartDateTime	datetimeoffset	Yes	
AnotherEndDateTime	datetimeoffset	Yes	

StatusId	uniqueidentifier	Yes	Foreign Key
IsActive	bit	Yes	
CreatedBy	uniqueidentifier	yes	Foreign Key
CreatedOn	datetimeoffset	yes	Default value as GETDATE()
UpdatedBy	uniqueidentifier	no	userId
UpdatedOn	datetimeoffset	no	Default value as GETDATE()

PTO.Quota

Column Name	Data Type	Required?	Description
Id	uniqueidentifier	yes	Primary Key
Name	nvarchar(30)	no	
Description	nvarchar(50)	no	
StartDateTime	datetimeoffset	yes	
EndDateTime	datetimeoffset	yes	
OriginalHours	decimal (4,1)	yes	
RemainingHours	decimal (4,1)	yes	
TeamId	uniqueidentifier	yes	Foreign Key
IsActive	Bit	yes	

CreatedBy	uniqueidentifier	yes	Foreign Key
CreatedOn	datetimeoffset	yes	Default value as `GETDATE()`
UpdatedBy	uniqueidentifier	no	Foreign Key
UpdatedOn	datetimeoffset	no	Default value as `GETDATE()`

Security.User

Column Name	Data Type	Required?	Description
Id	uniqueidentifier	yes	Primary Key
FirstName	nvarchar(30)	yes	
LastName	nvarchar(30)	yes	
NTLogin	nvarchar(12)	yes	
EmailAddress	nvarchar(50)	yes	
RoleId	uniqueidentifier	yes	Foreign Key
ReportToUserId	uniqueidentifier	yes	Foreign Key
LocationId	uniqueidentifier	yes	Foreign Key
TeamId	uniqueidentifier	yes	Foreign Key to Security.Team
IsActive	bit	yes	
CreatedBy	uniqueidentifier	yes	Foreign Key
CreatedOn	datetimeoffset	yes	Default value as `GETDATE()`
UpdatedBy	uniqueidentifier	no	Foreign Key
UpdatedOn	datetimeoffset	no	Default value as `GETDATE()`

Role :

1. Manager
2. Developer 1
3. Developer 2

Security.Role

Column Name	Data Type	Required?	Description
Id	uniqueidentifier	yes	Primary Key
Name	nvarchar(30)	yes	
Description	nvarchar(50)	no	
IsLeadership	bit	yes	
IsActive	bit	yes	
CreatedBy	uniqueidentifier	yes	Foreign Key
CreatedOn	datetimeoffset	yes	Default value as `GETDATE()`
UpdatedBy	uniqueidentifier	no	Foreign Key
UpdatedOn	datetimeoffset	no	Default value as `GETDATE()`

Teams :

1. Help Desk 8am-9pm
2. OnBoarding / Recovery (OBR) 9am-6pm
3. Leadership / Admin 8am-5pm
4. Sales Order Entry (SOE) 8am-6pm

MaxShiftSlideHours : 3 Hours

Security.Team

Column Name	Data Type	Required?	Description
Id	uniqueidentifier	yes	Primary Key
Name	nvarchar(30)	yes	
Description	nvarchar(50)	no	
IsLeadership	bit	yes	
MaxShiftSlideHours	decimal(4,1)	yes	
ShiftStartTimeLimit	decimal(4,1)	yes	
ShiftEndTimeLimit	decimal(4,1)	yes	
IsActive	bit	yes	
CreatedBy	uniqueidentifier	yes	Foreign Key
CreatedOn	datetimeoffset	yes	Default value as `GETDATE()`
UpdatedBy	uniqueidentifier	no	Foreign Key
UpdatedOn	datetimeoffset	no	Default value as `GETDATE()`

Location :

1. Nashville
2. Schaumburg

Config.Location

Column Name	Data Type	Required?	Description
Id	uniqueidentifier	yes	Primary Key
Name	nvarchar(30)	yes	
Description	nvarchar(50)	no	
IsActive	bit	yes	
CreatedBy	uniqueidentifier	yes	Foreign Key
CreatedOn	datetimeoffset	yes	Default value as `GETDATE()`
UpdatedBy	uniqueidentifier	no	Foreign Key
UpdatedOn	datetimeoffset	no	Default value as `GETDATE()`

Config.RequestType

Time-off Category :

1. Vacation
2. Personnel Swap
3. Flex Time
4. Float
5. Bereavement
6. Shift Slide
7. Jury Duty

(Need to prepare work flow for each category)

Column Name	Data Type	Required?	Description
Id	uniqueidentifier	yes	Primary Key
Name	nvarchar(30)	yes	
Description	nvarchar(50)	no	
IsActive	bit	yes	
CreatedBy	uniqueidentifier	yes	Foreign Key
CreatedOn	datetimeoffset	yes	Default value as `GETDATE()`
UpdatedBy	uniqueidentifier	no	Foreign Key
UpdatedOn	datetimeoffset	no	Default value as `GETDATE()`

Config.FlexType

1. Shift Slide
2. Pre-Arranged Shift Slide
3. Shift Swap
4. Self-Shift Swap
5. Shift Swap Request

(Need to prepare work flow for each category)

Column Name	Data Type	Required?	Description
Id	uniqueidentifier	yes	Primary Key
Name	nvarchar(30)	yes	
Description	nvarchar(50)	no	
IsActive	bit	yes	
CreatedBy	uniqueidentifier	yes	Foreign Key
CreatedOn	datetimeoffset	yes	Default value as `GETDATE()`
UpdatedBy	uniqueidentifier	no	Foreign Key
UpdatedOn	datetimeoffset	no	Default value as `GETDATE()`

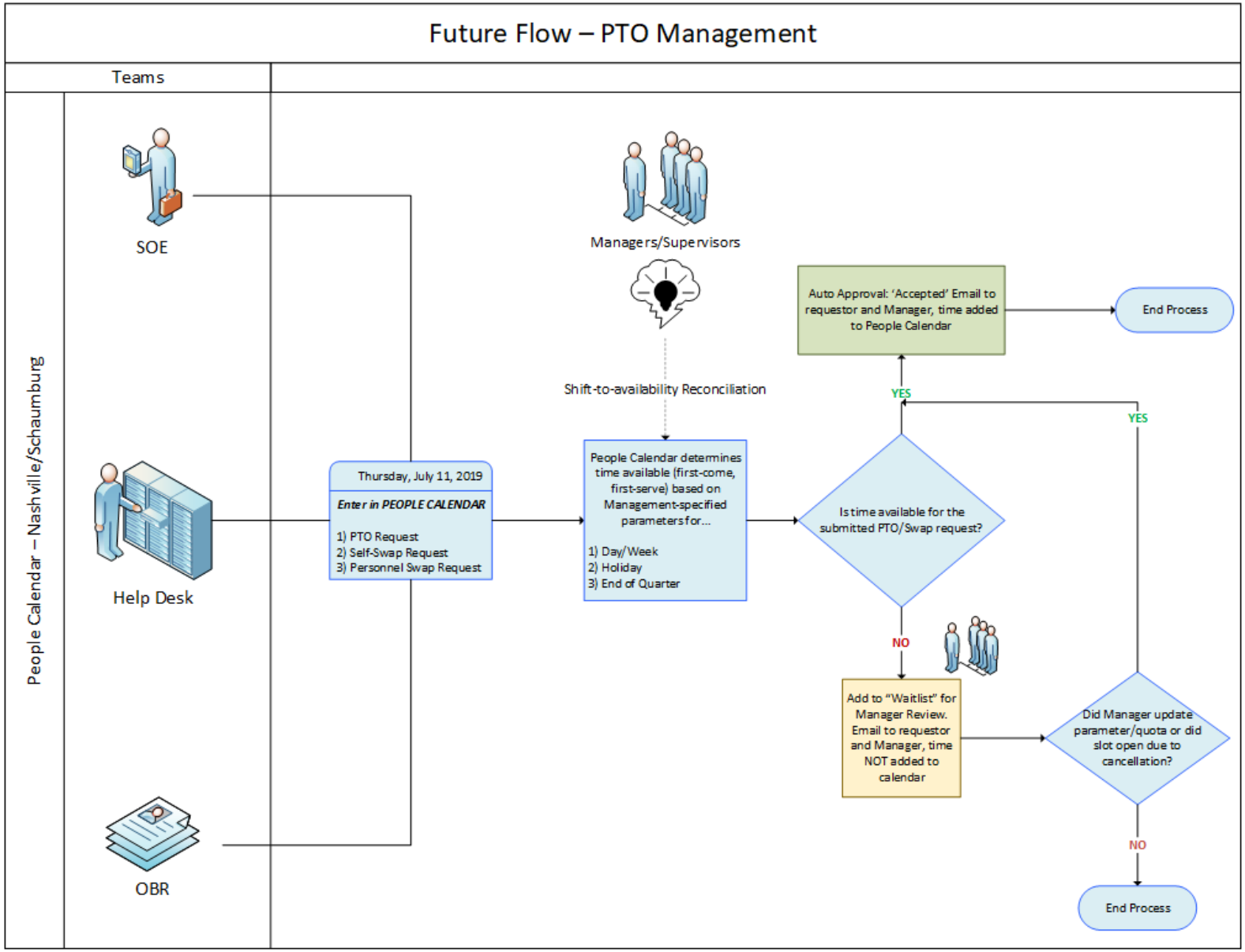
Status :

1. WaitList
2. Approved
3. Cancelled
4. Pending
5. Declined

Config.Status

Column Name	Data Type	Required?	Description
Id	uniqueidentifier	yes	Primary Key
Name	nvarchar(30)	yes	
Description	nvarchar(50)	no	
IsActive	bit	yes	
CreatedBy	uniqueidentifier	yes	Foreign Key
CreatedOn	datetimeoffset	yes	Default value as `GETDATE()`
UpdatedBy	uniqueidentifier	no	Foreign Key
UpdatedOn	datetimeoffset	no	Default value as `GETDATE()`

Expected Process Work Flow :





Schedule Flexibility Options

Schedule Flexibility options can help you balance your work and personal life



- **Shift Swap** (Trade a day or a shift with a co-worker)
  - Day Trade Example: Coworker works Tue-Sat and you work Sun-Thur. You can trade your Mon for their Saturday. You must also work your coworkers shift on the day you trade. If they had an 11-8, you must work 11-8 on the day you trade. The same is true on the day they work for you
  - Shift Trade Example: Coworker works 7 am – 4 pm, you work 11 am – 8 pm...you can take their 7 am – 4 pm shift as long as they take your 11 am – 8 pm shift on the same day

*These options may be used an unlimited number of times and must be communicated to both leaders, via email, at least 24 hours in advance. Both agents must acknowledge acceptance of the swap in the email chain. Swaps must occur during the payroll week, running Sunday- Saturday

- **Self-Shift Swap** (Swap your own off day with a work day)
  - Example: You are normally off Sun/Mon, but you'd like to be off Sat instead, you can swap your Sun or Mon for Sat.

*This option may be used twice per month and requires prior approval. Your leader must be contacted at least 24 hours in advance. Blackout dates: 21st of each month, all company holidays, national holidays, Easter and the day before and after a holiday. Swaps must occur during the payroll week, running Sunday- Saturday. Self Swaps are subject to business needs at the discretion of leadership



- **Shift Slide** (shift your start and end time)
  - Shift Slide allows you to meet personal needs that arise due to unforeseeable events by moving your schedule up to 3 hours forward or backward
  - You may also slide up to 30 minutes of your lunch break for a shift slide (i.e. you normally work 11 am – 8 pm, but you need to start at 11:30 am. If approved, you can start at 11:30 am and cut 30 minutes out of your 60 minute lunch and still leave at 8 pm)

*This option may be used three in a calendar month. Shift slides must fall within the operating hours of Sales Operations (example...you can't work until 10 pm). You must notify your leader at least 30 minutes in advance of the start time of your original shift

- **Pre-Arranged Shift Slide** (shift your start and end time in advance)
  - Works the same as shift slide (shown above), but scheduled in advance

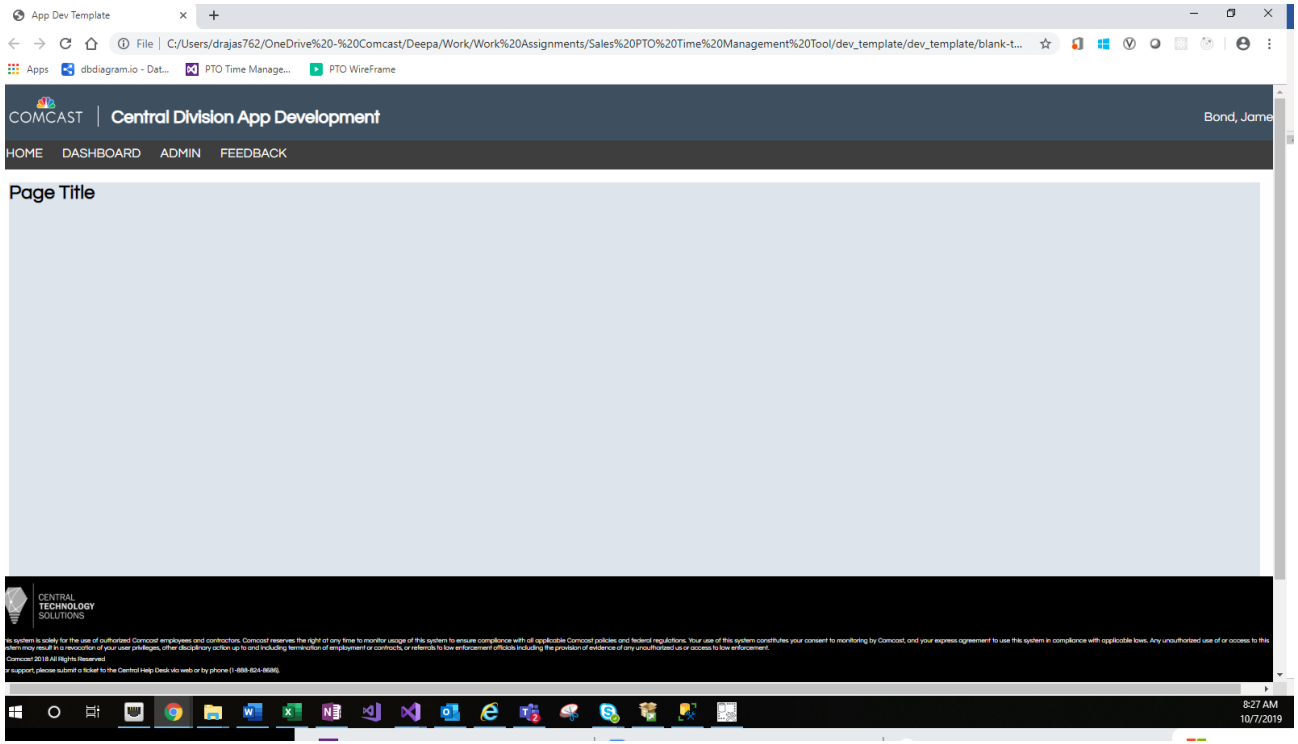
*This option may be used twice in a calendar month. Pre-arranged shift slides must fall within the operating hours of Sales Operations (example...you can't work until 10 pm). You must notify your leader at least 72 hours in advance of the start time of your original shift

Flexibility options can only be used within the frequency shown. More than one flexibility option cannot be exercised on a single day. For more information about restrictions and process, please see your leader. Policy is subject to change, based on business needs

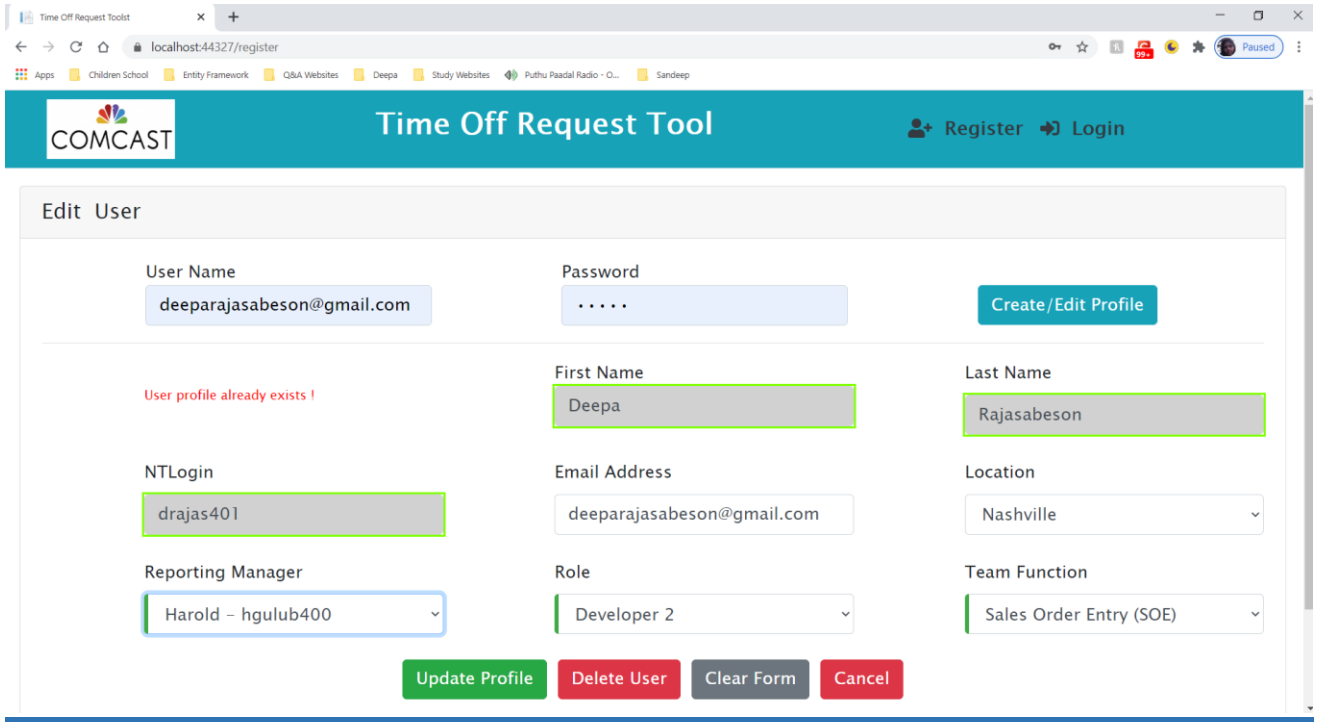
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User Interface Features

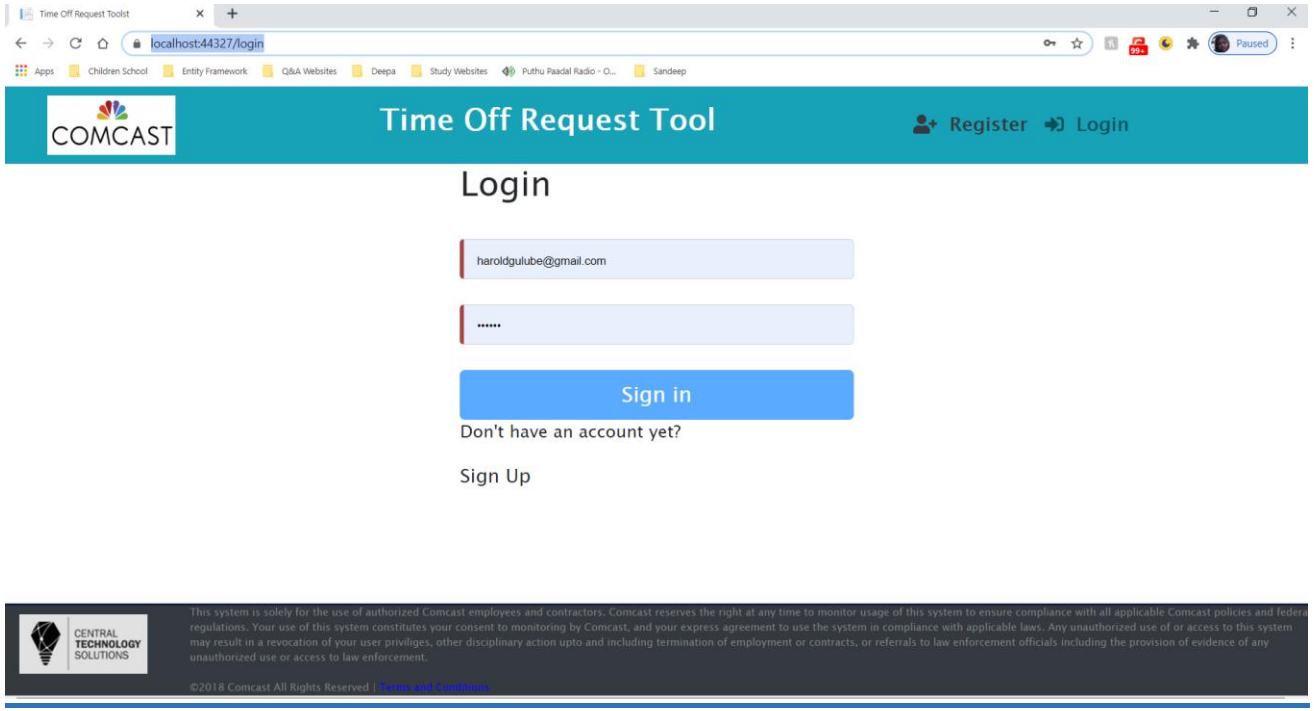
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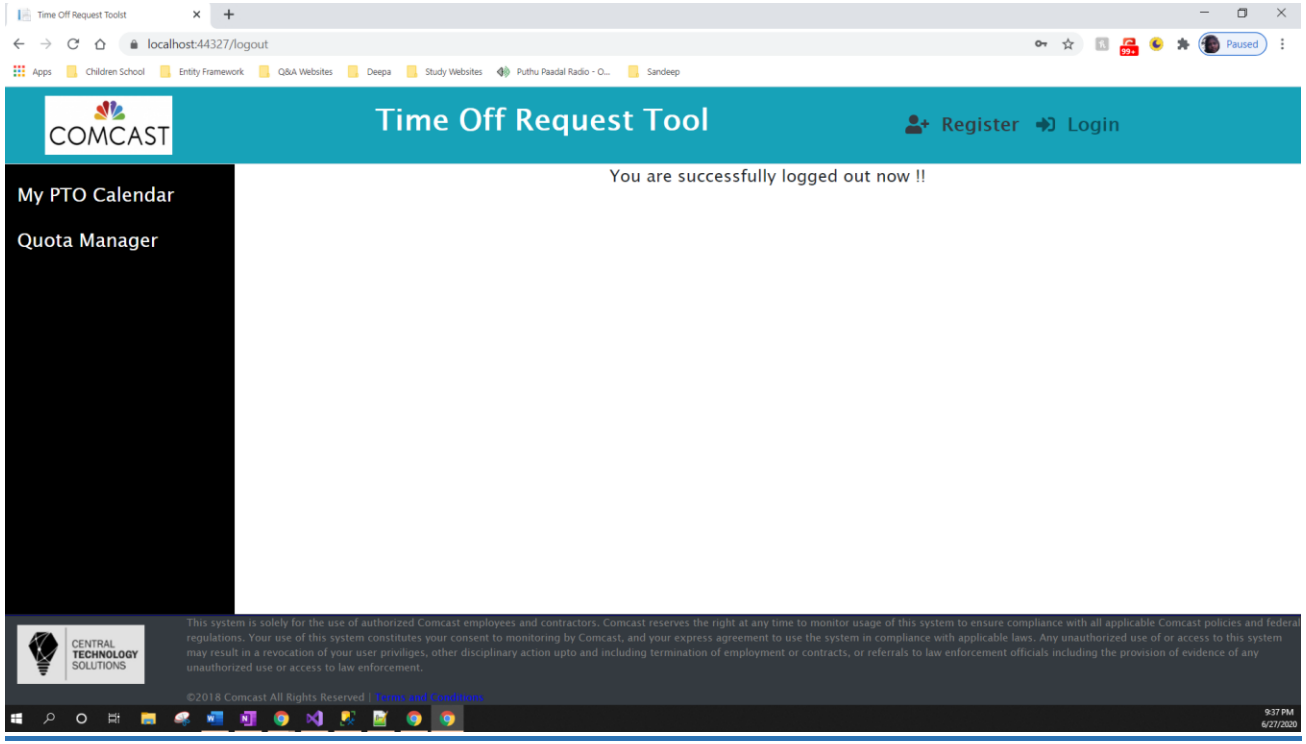


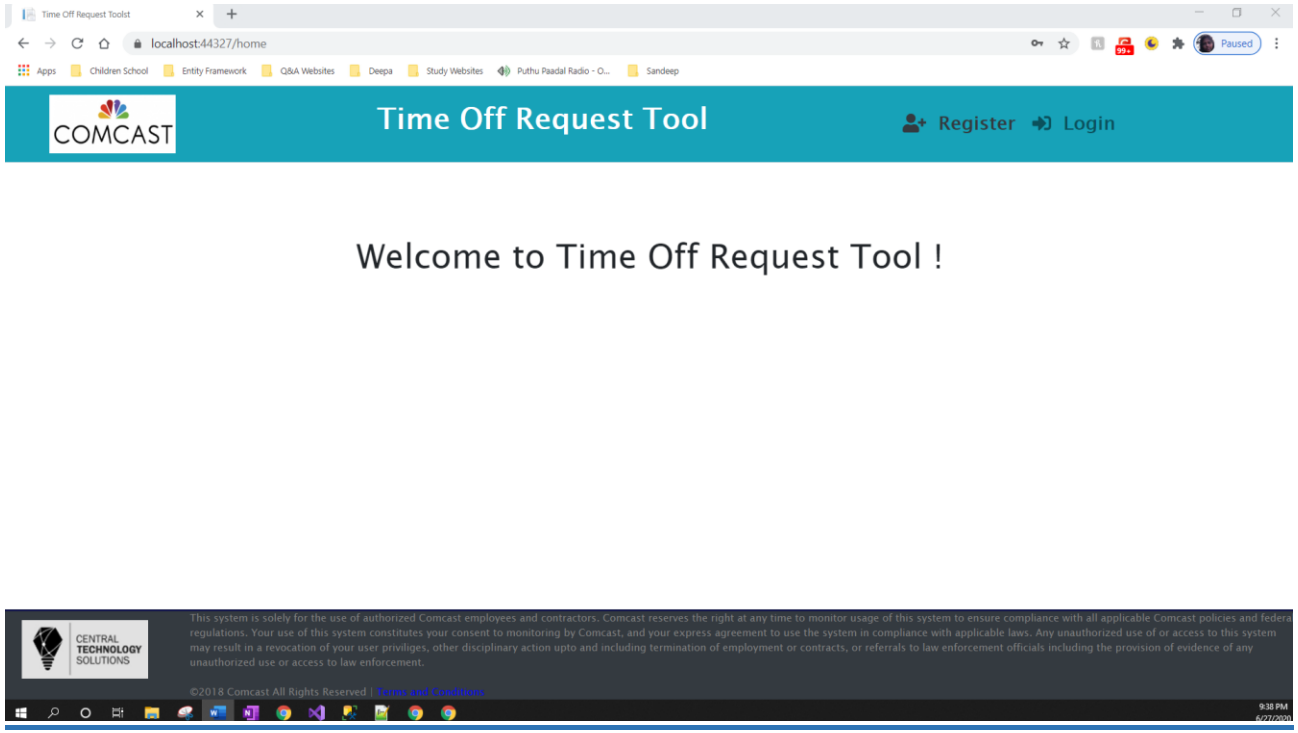
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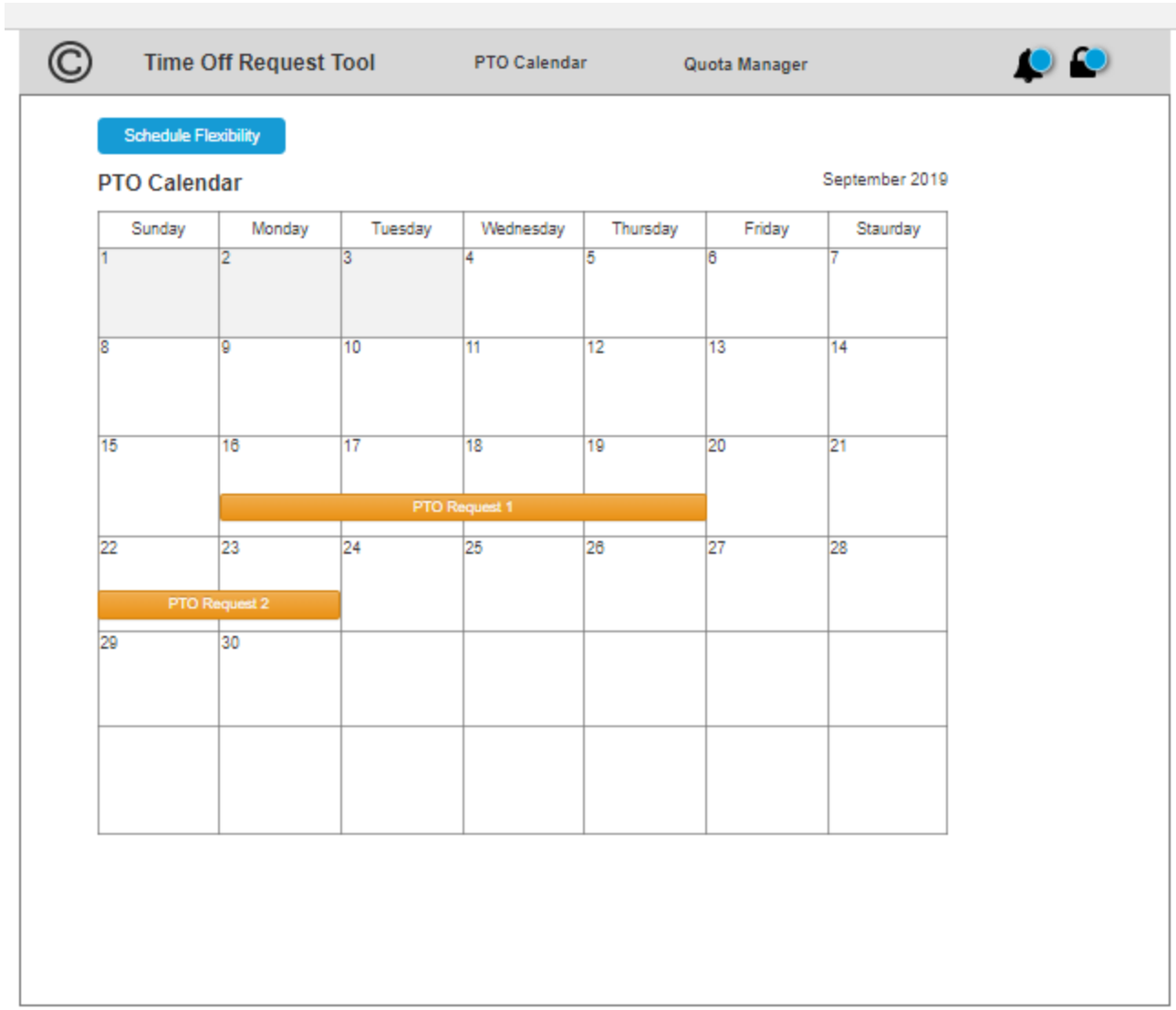


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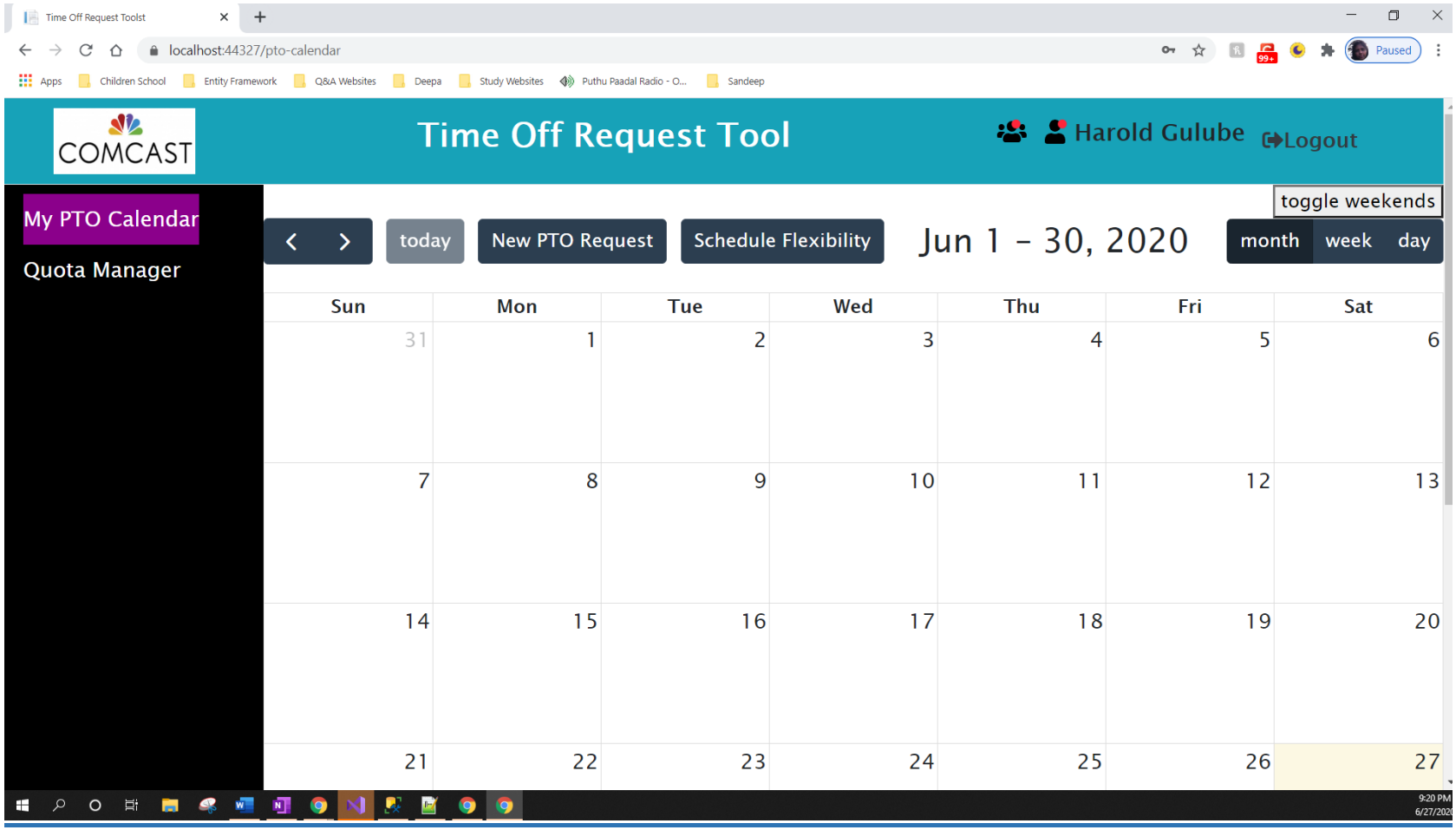

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Entity Framework
Q&A Websites
Deepa
Study Websites
Puthu Paadal Radio - O...
Sandeep
COMCAST
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Register
Login
Login
haroldgulube@gmail.com
Sign in
Don't have an account yet?
Sign Up
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Children School
Entity Framework
Q&A Websites
Deepa
Study Websites
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Time Off Request Tool
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Login
My PTO Calendar
Quota Manager
You are successfully logged out now !!
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Study Websites
Puthu Paadal Radio - O...
Sandeep
COMCAST
Time Off Request Tool
Register
Login
Welcome to Time Off Request Tool !
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6/27/2020


Time Off Request Tool
PTO Calendar
Quota Manager
Schedule Flexibility
PTO Calendar
September 2019
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Staurday
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PTO Request 1
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PTO Request 2
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Application Image (PTO Calendar) :



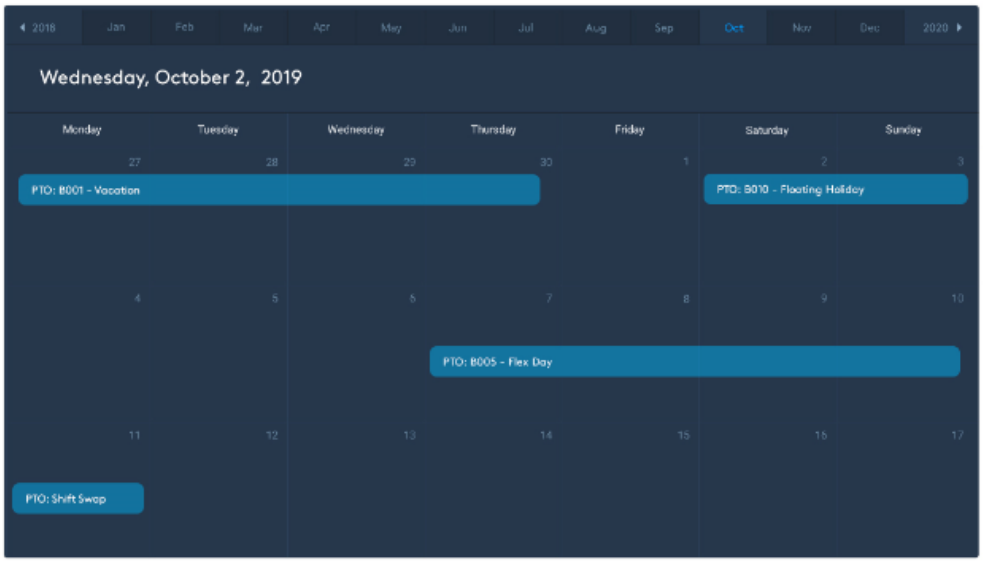
COMCAST | TimeOff Request Tool

6 Annie, Newton

My PTO Calendar Quota Manager

My PTO Calendar

Change Your Schedule

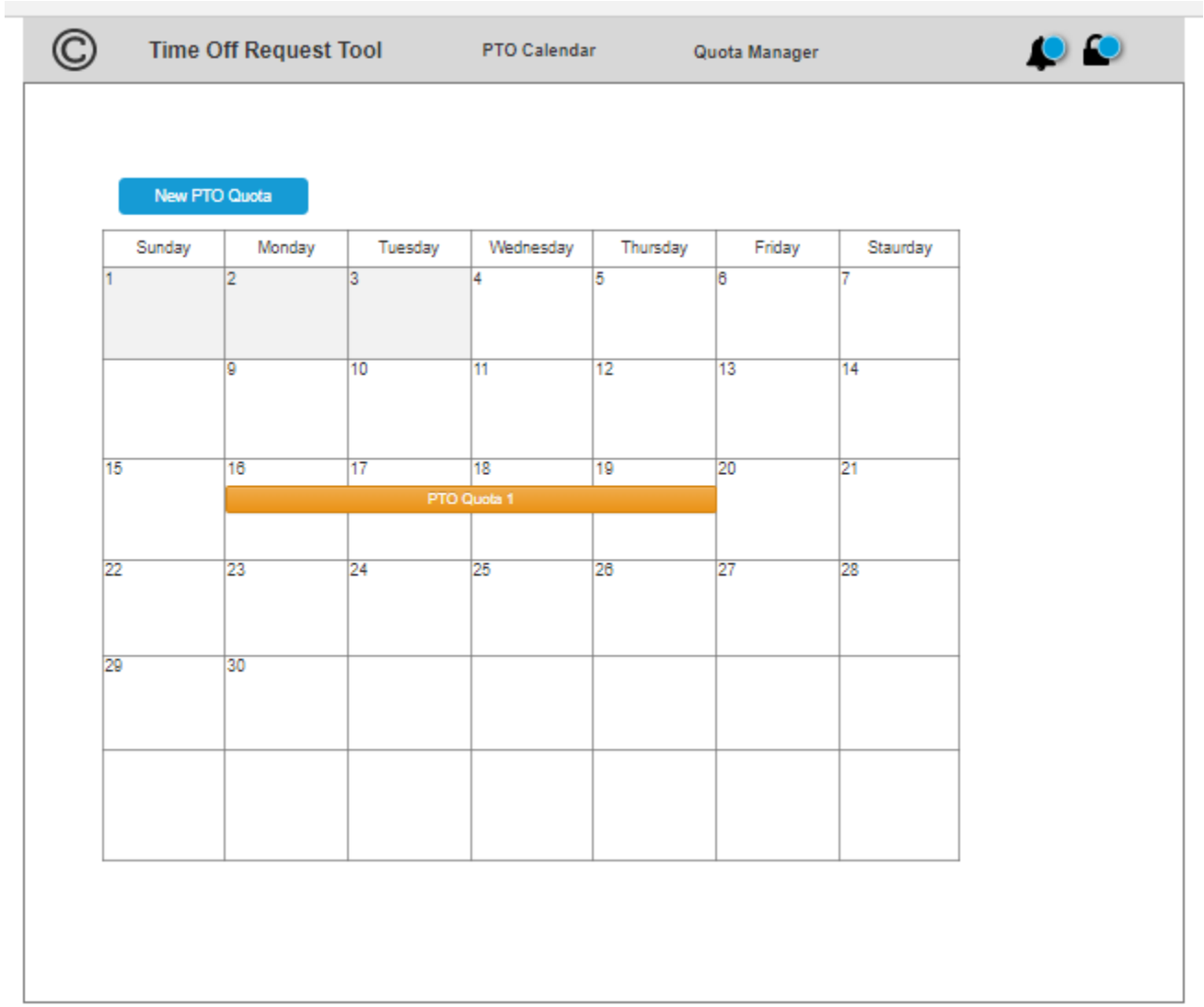


CENTRAL TECHNOLOGY SOLUTIONS

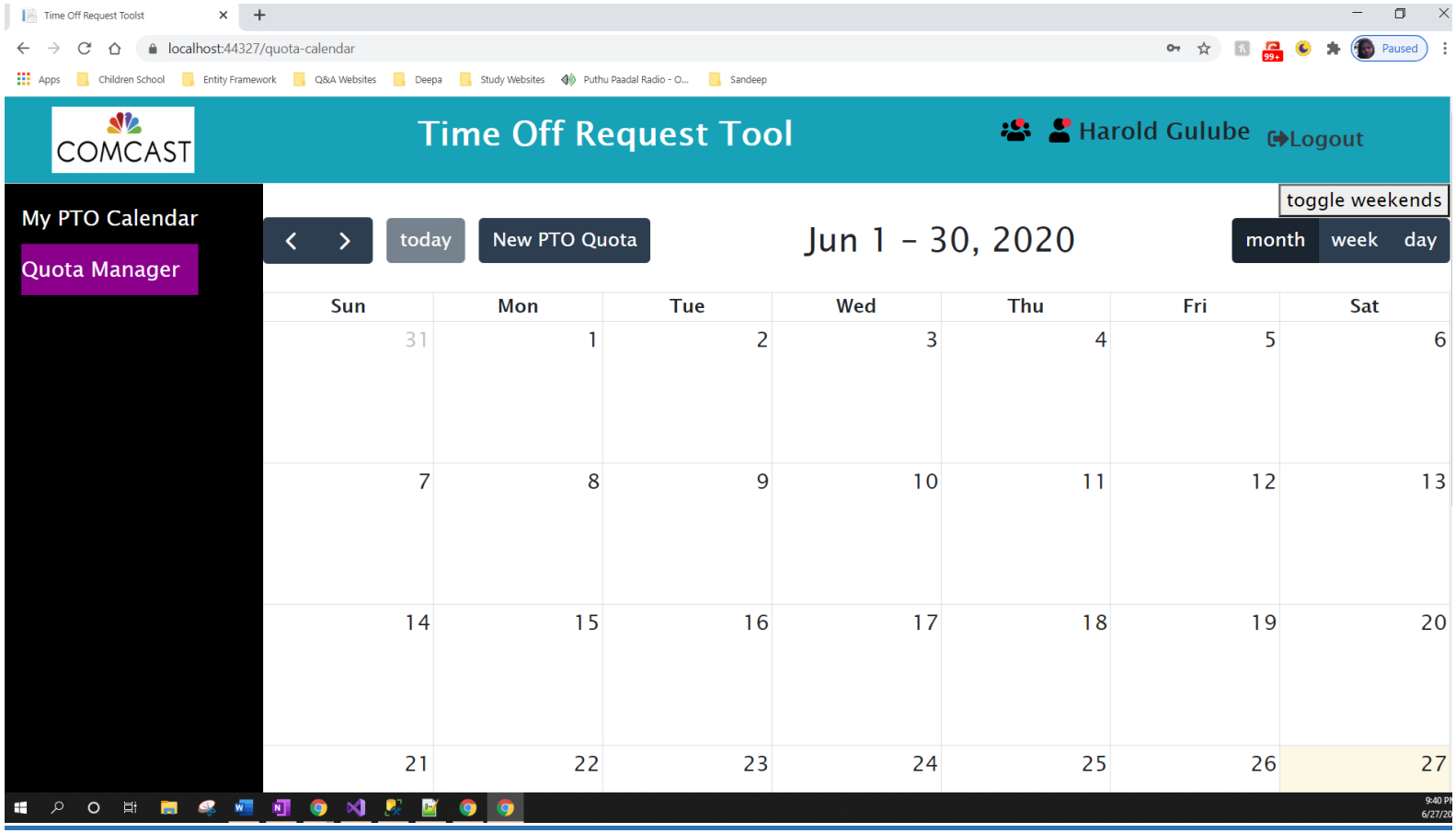
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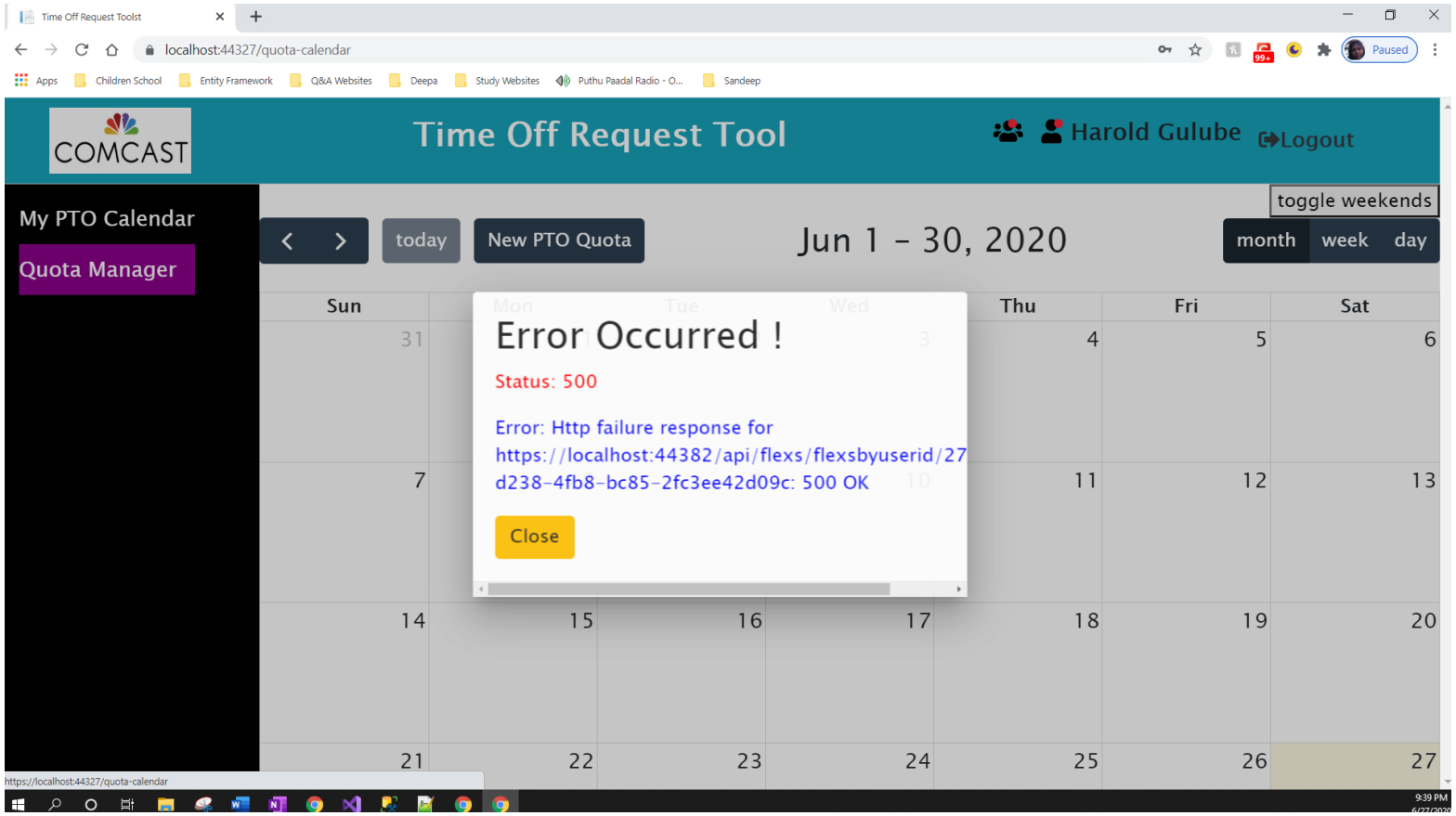
Quota Manager :



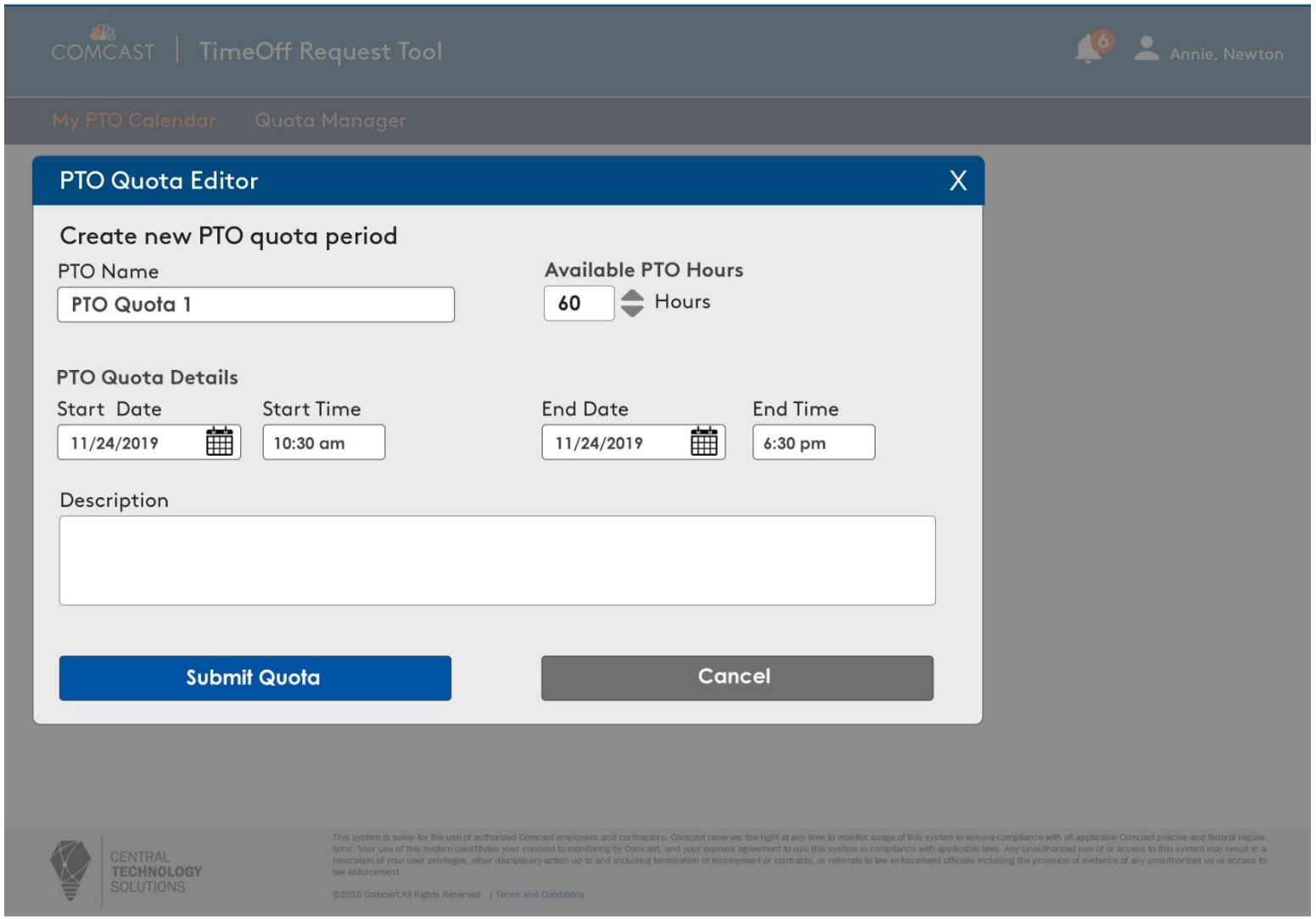
Application Image (Quota Calendar) :



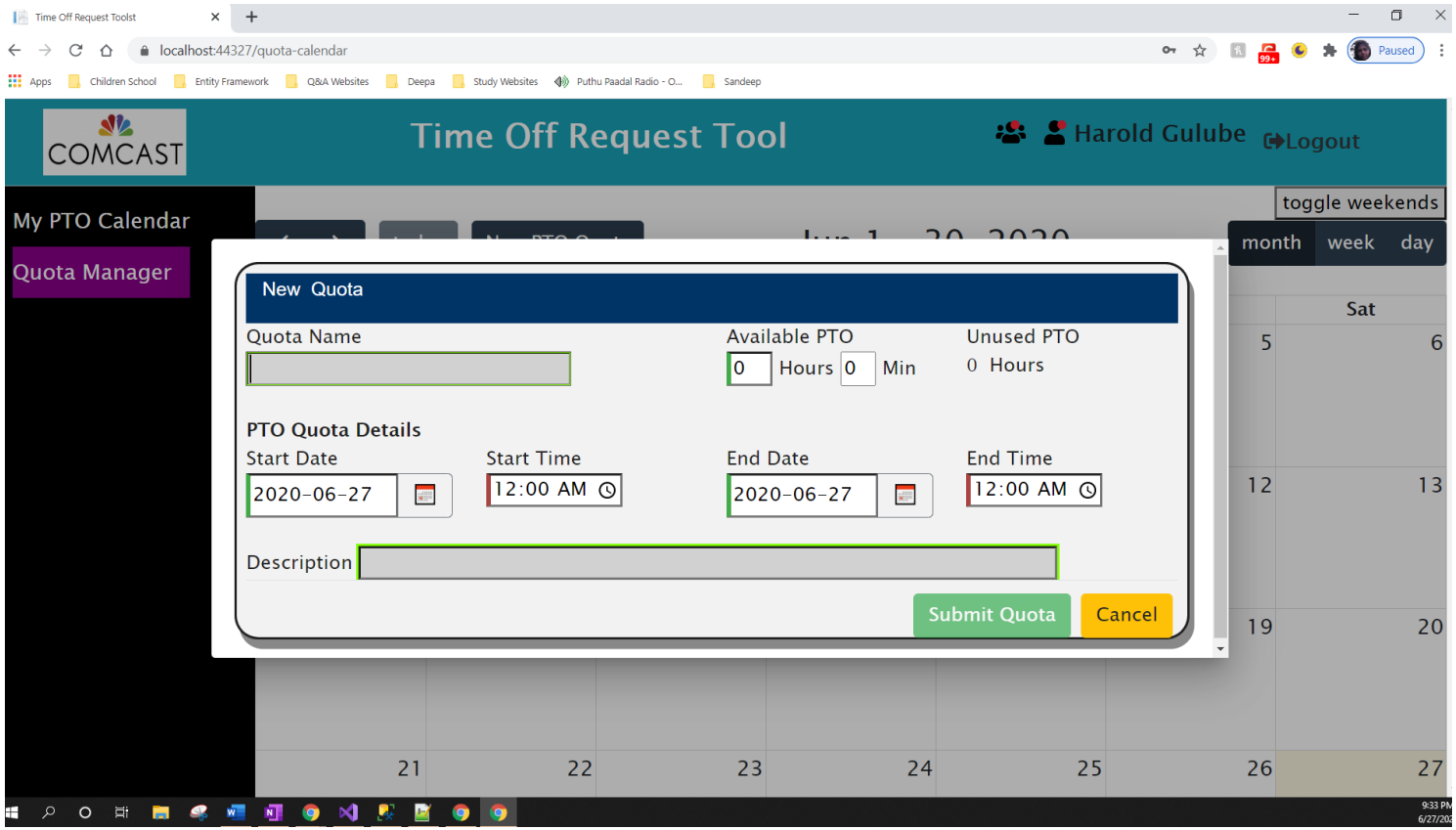
Application Image (Error Display) :



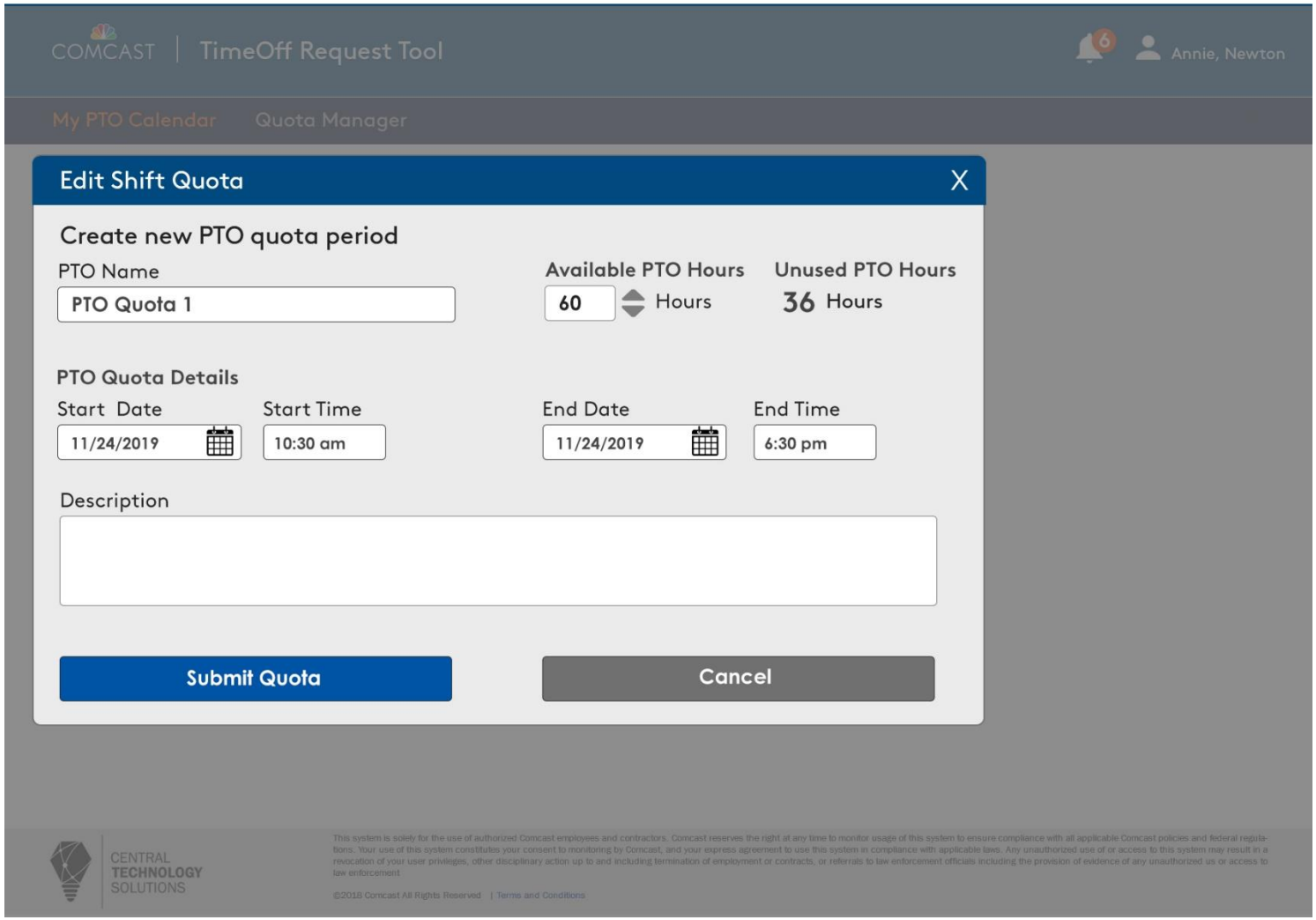
PTO Quota Editor (Manager View) :


COMCAST | TimeOff Request Tool
6
Annie, Newton
My PTO Calendar
Quota Manager
PTO Quota Editor
X
Create new PTO quota period
PTO Name
PTO Quota 1
Available PTO Hours
60
Hours
PTO Quota Details
Start Date
11/24/2019
Start Time
10:30 am
End Date
11/24/2019
End Time
6:30 pm
Description
Submit Quota
Cancel
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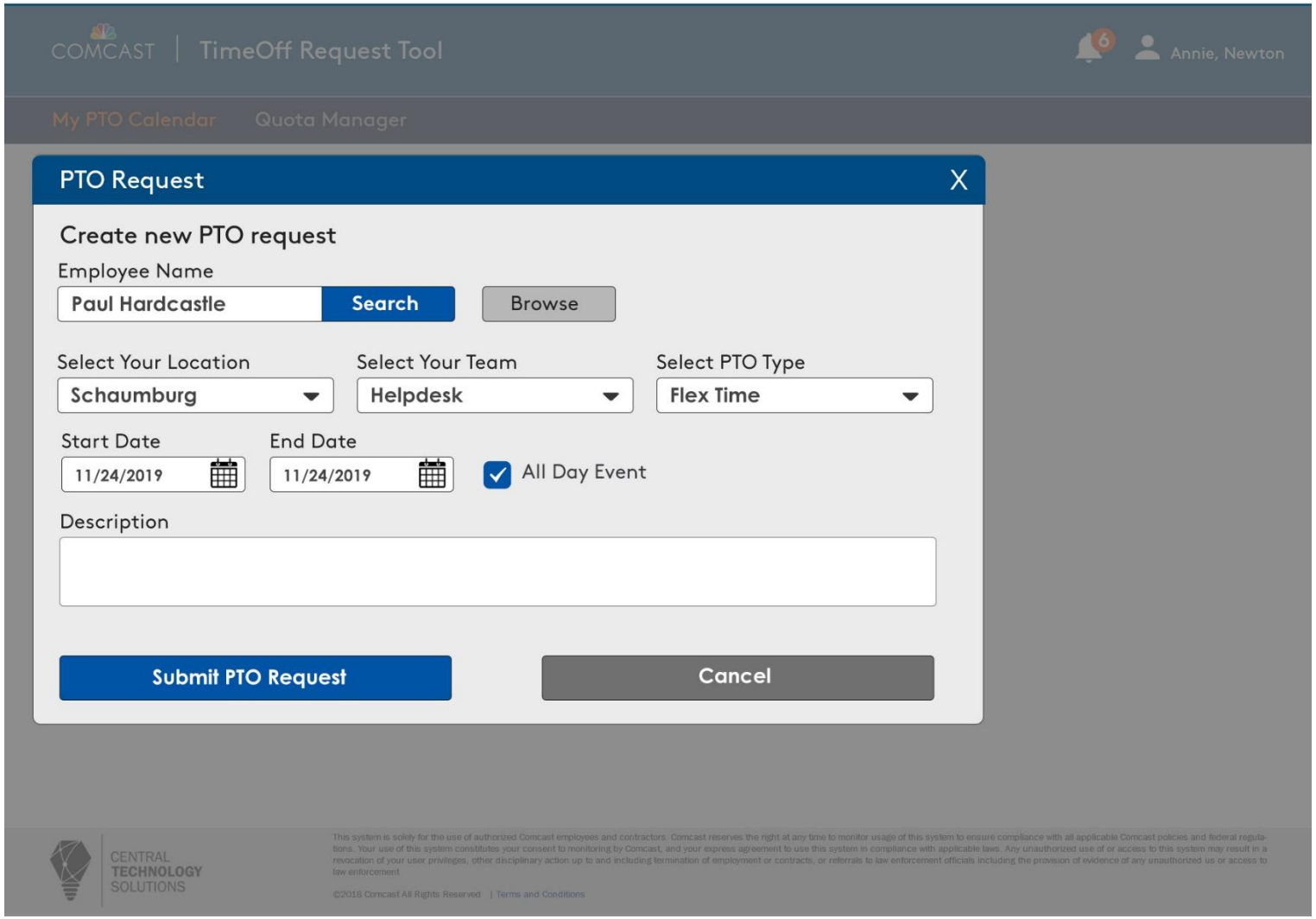
Application Image (Quota) :



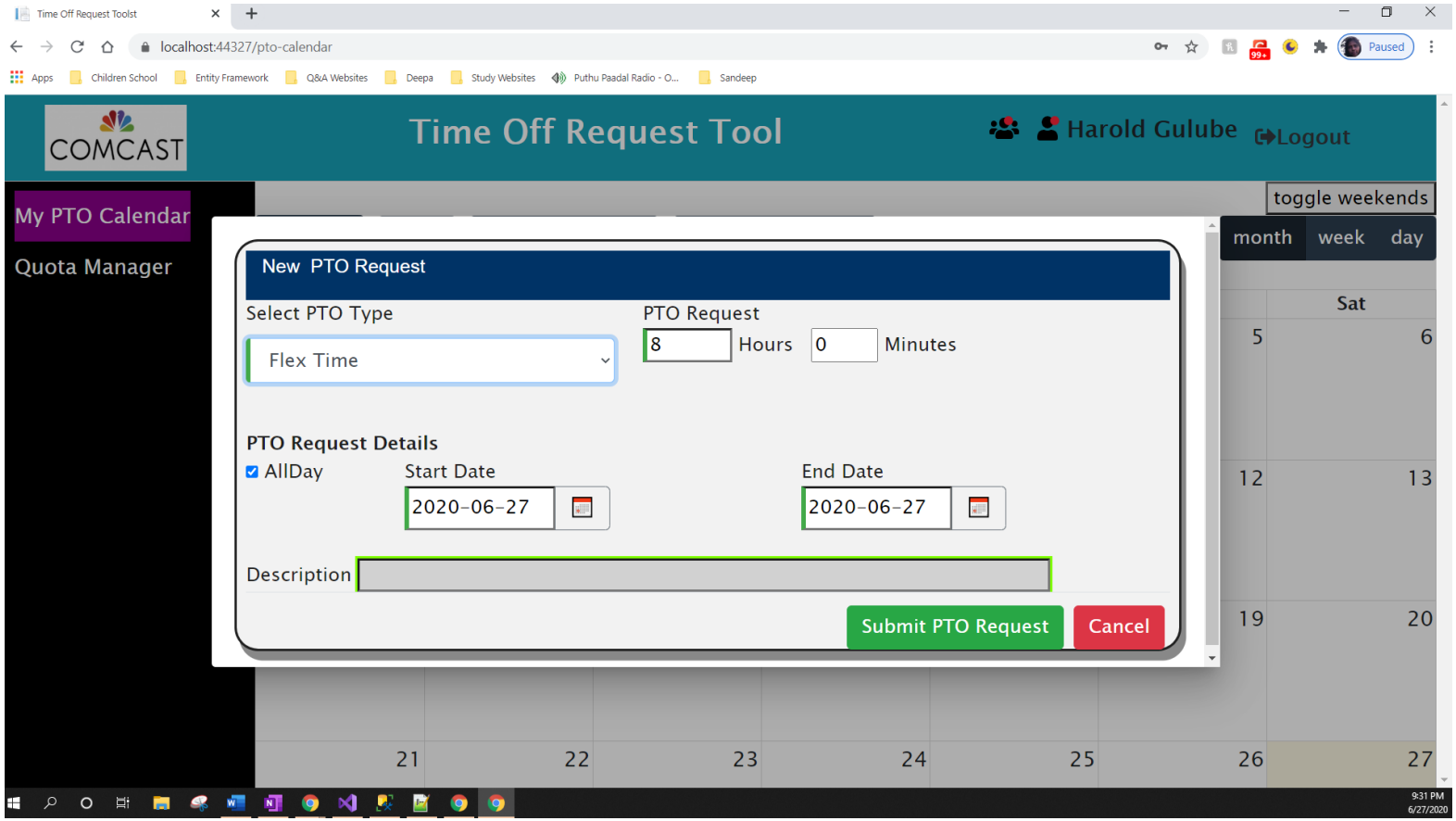
Edit Shift Quota (Manager View) :



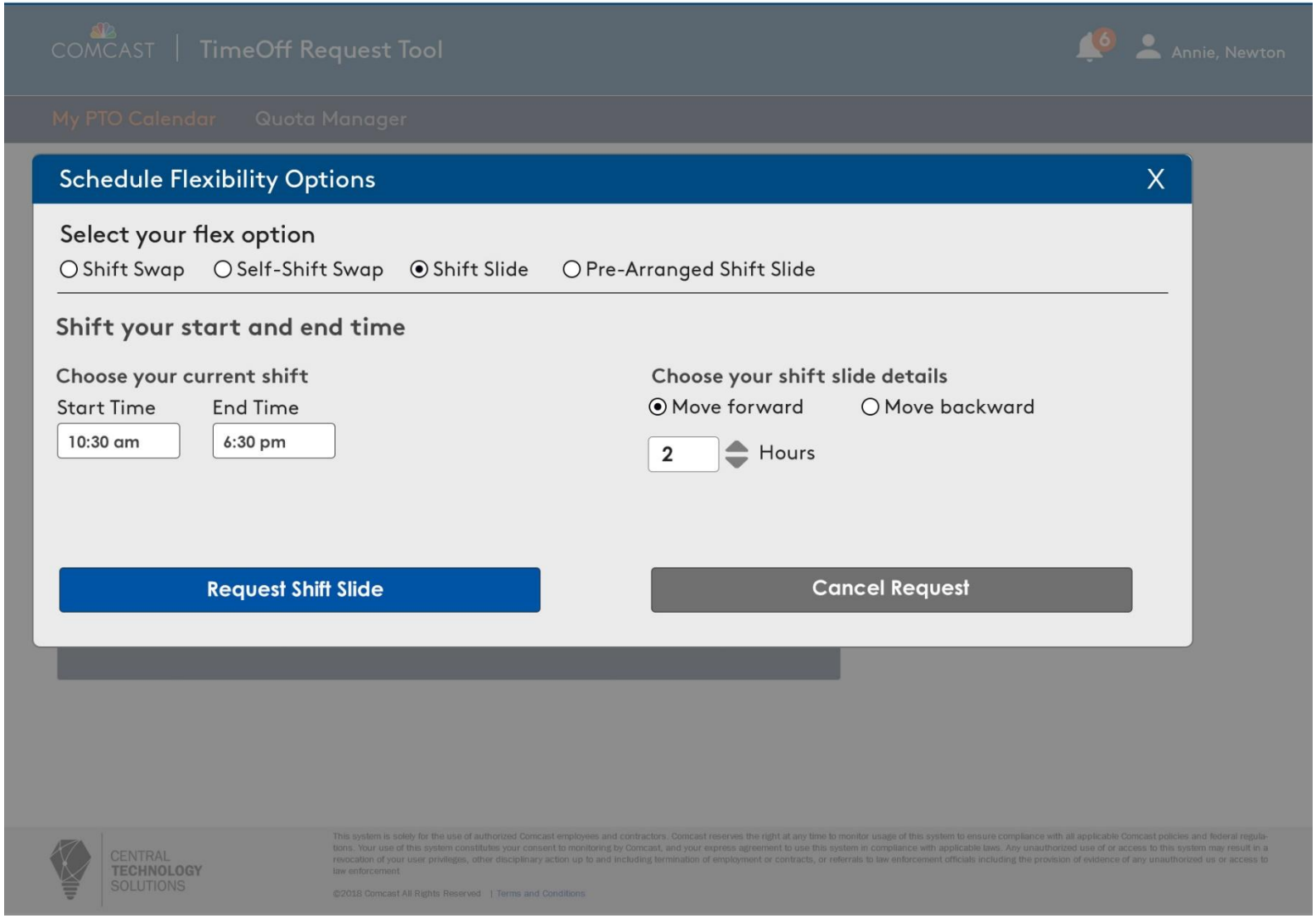
Create PTO Request Form (Employee View) :


COMCAST | TimeOff Request Tool
6
Annie, Newton
My PTO Calendar
Quota Manager
PTO Request
X
Create new PTO request
Employee Name
Paul Hardcastle
Search
Browse
Select Your Location
Schaumburg
Select Your Team
Helpdesk
Select PTO Type
Flex Time
Start Date
11/24/2019
End Date
11/24/2019
All Day Event
Description
Submit PTO Request
Cancel
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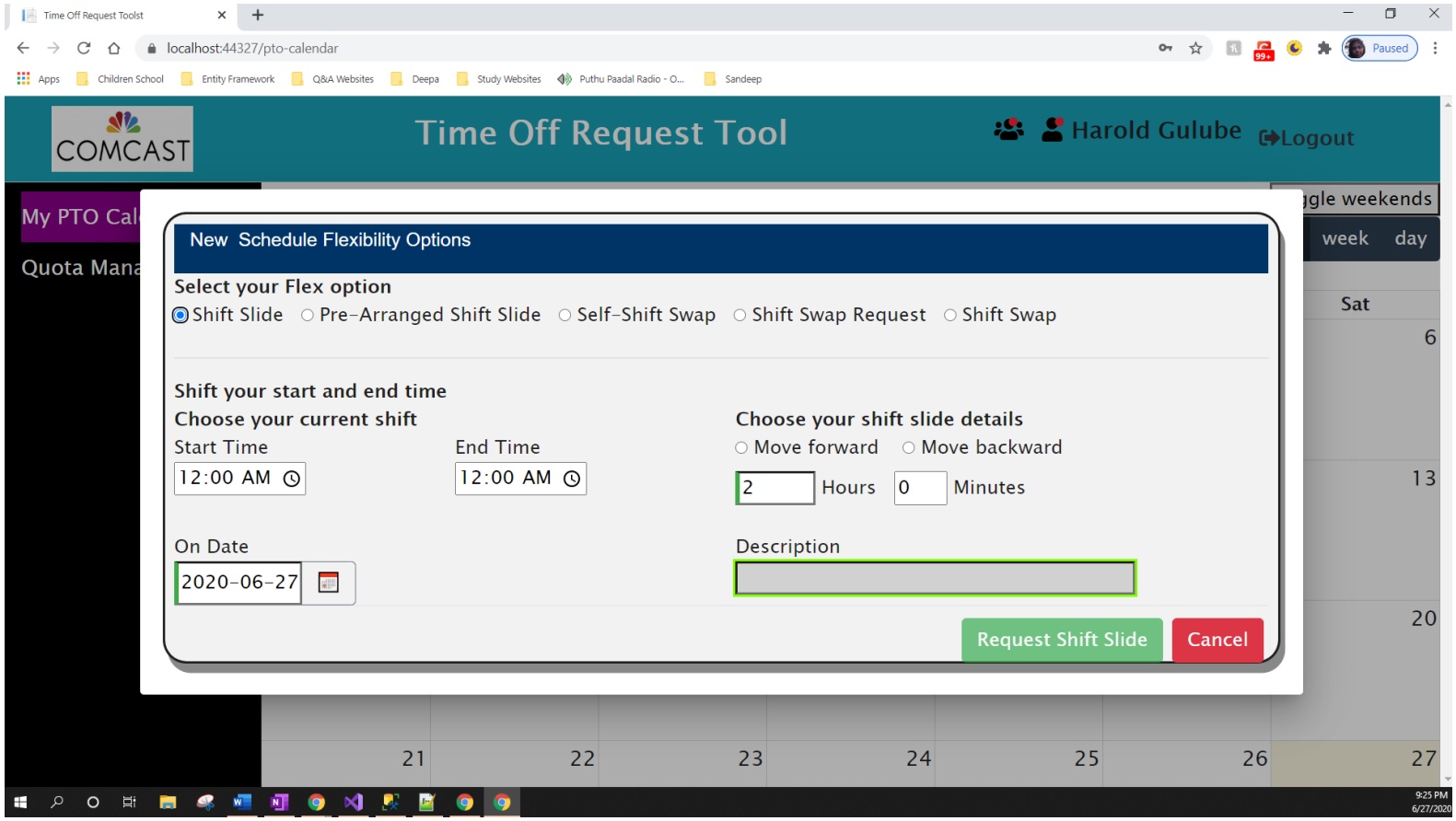
Application Image (PTO Request) :



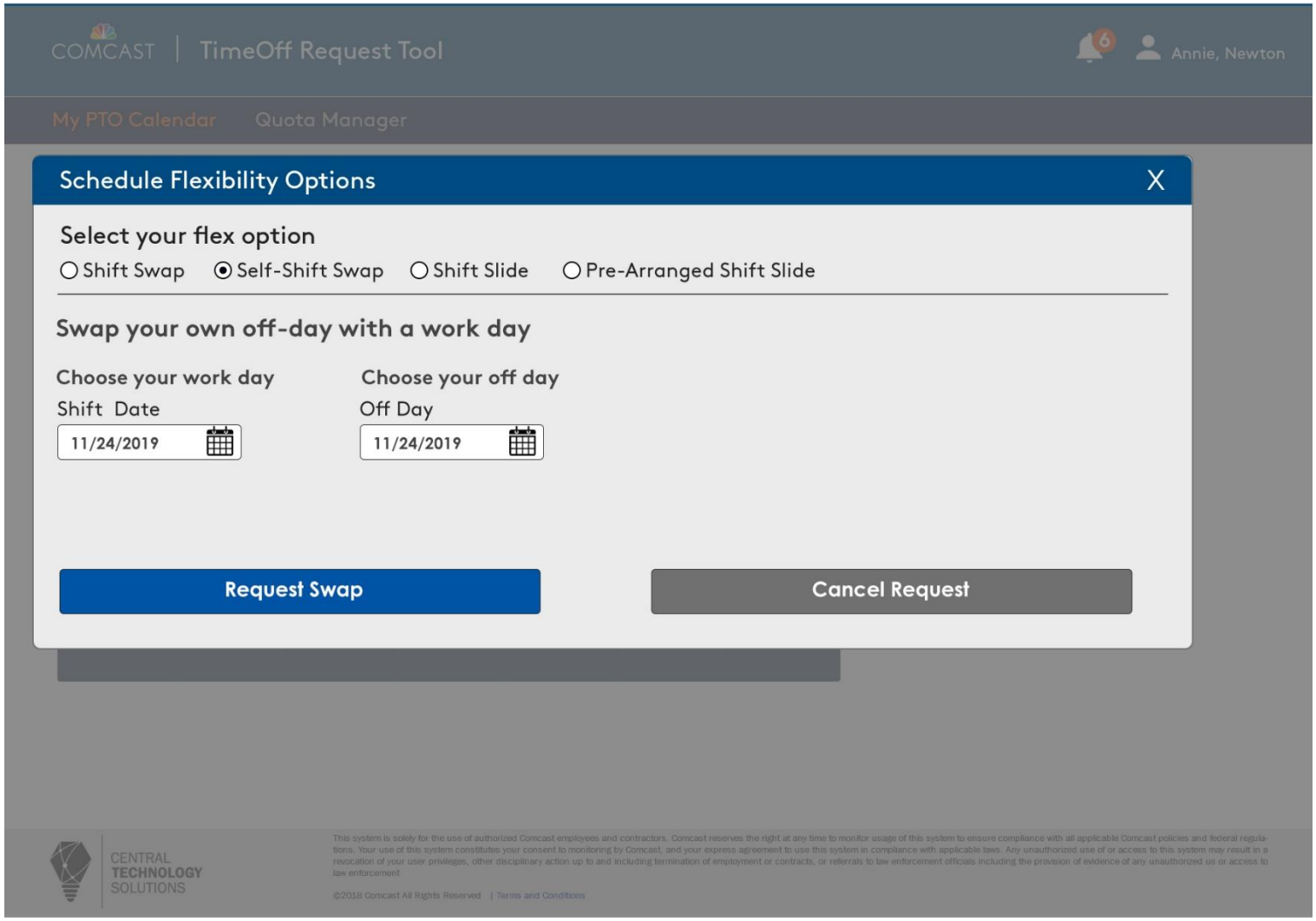
Shift Slide - Schedule Flex (Employee View) :


COMCAST | TimeOff Request Tool
6
Annie, Newton
My PTO Calendar
Quota Manager
Schedule Flexibility Options
X
Select your flex option
Shift Swap
Self-Shift Swap
Shift Slide
Pre-Arranged Shift Slide
Shift your start and end time
Choose your current shift
Start Time
End Time
10:30 am
6:30 pm
Choose your shift slide details
Move forward
Move backward
2
Hours
Request Shift Slide
Cancel Request
CENTRAL
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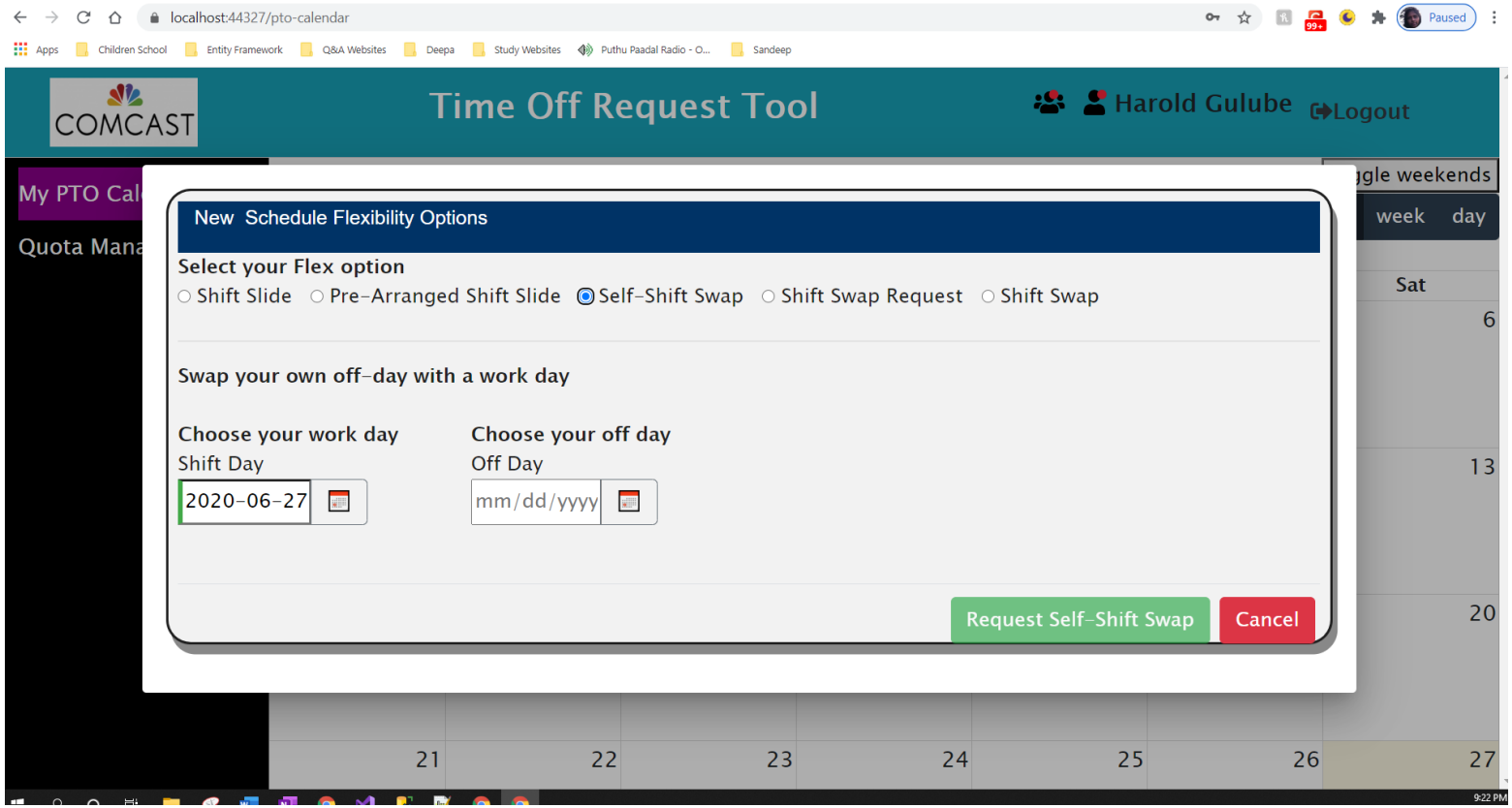
Application Image (Shift Slide) :



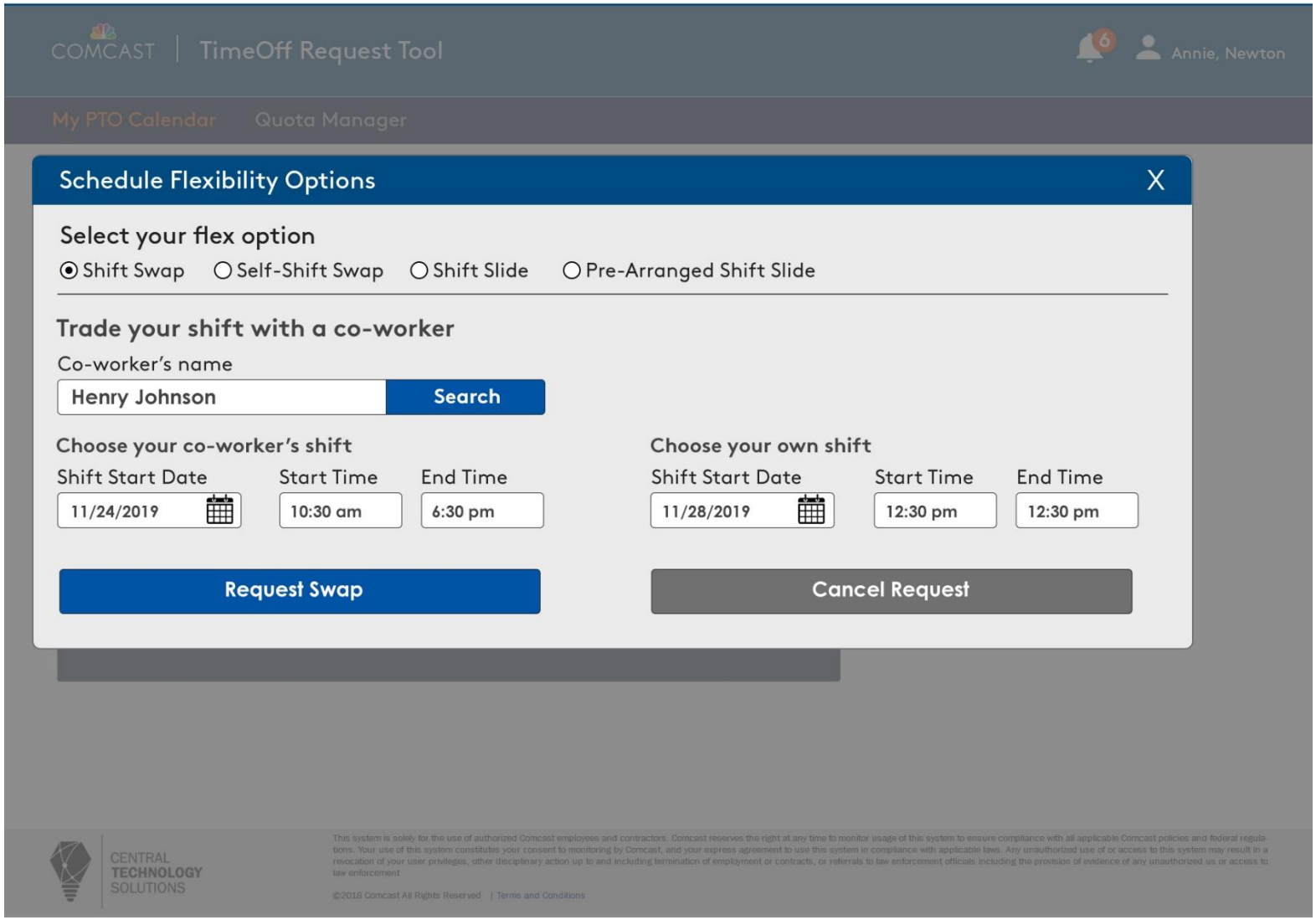
Self-Shift Swap - Schedule Flex (Employee View) :



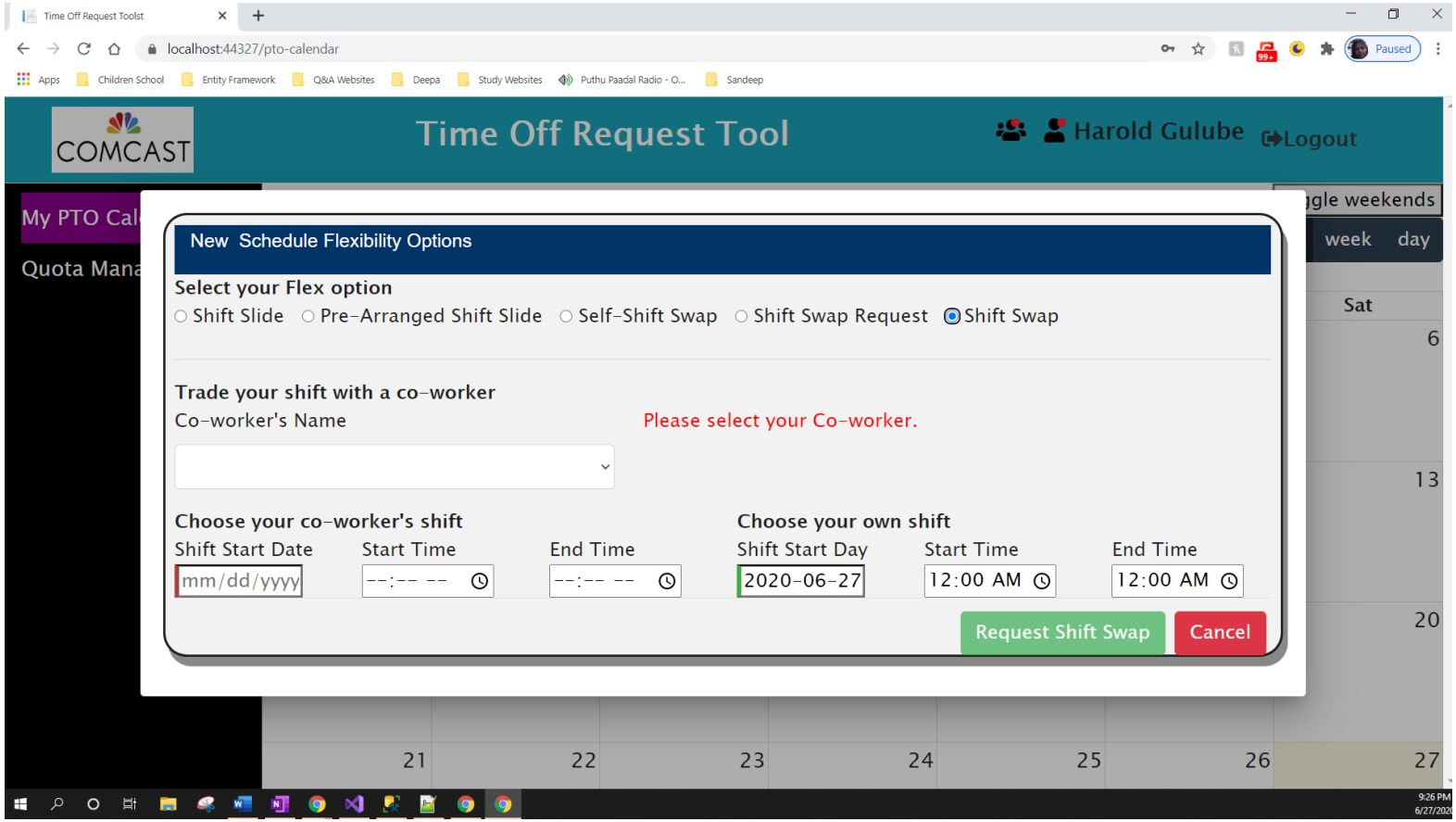
Application Image (Self-Shift Slide) :



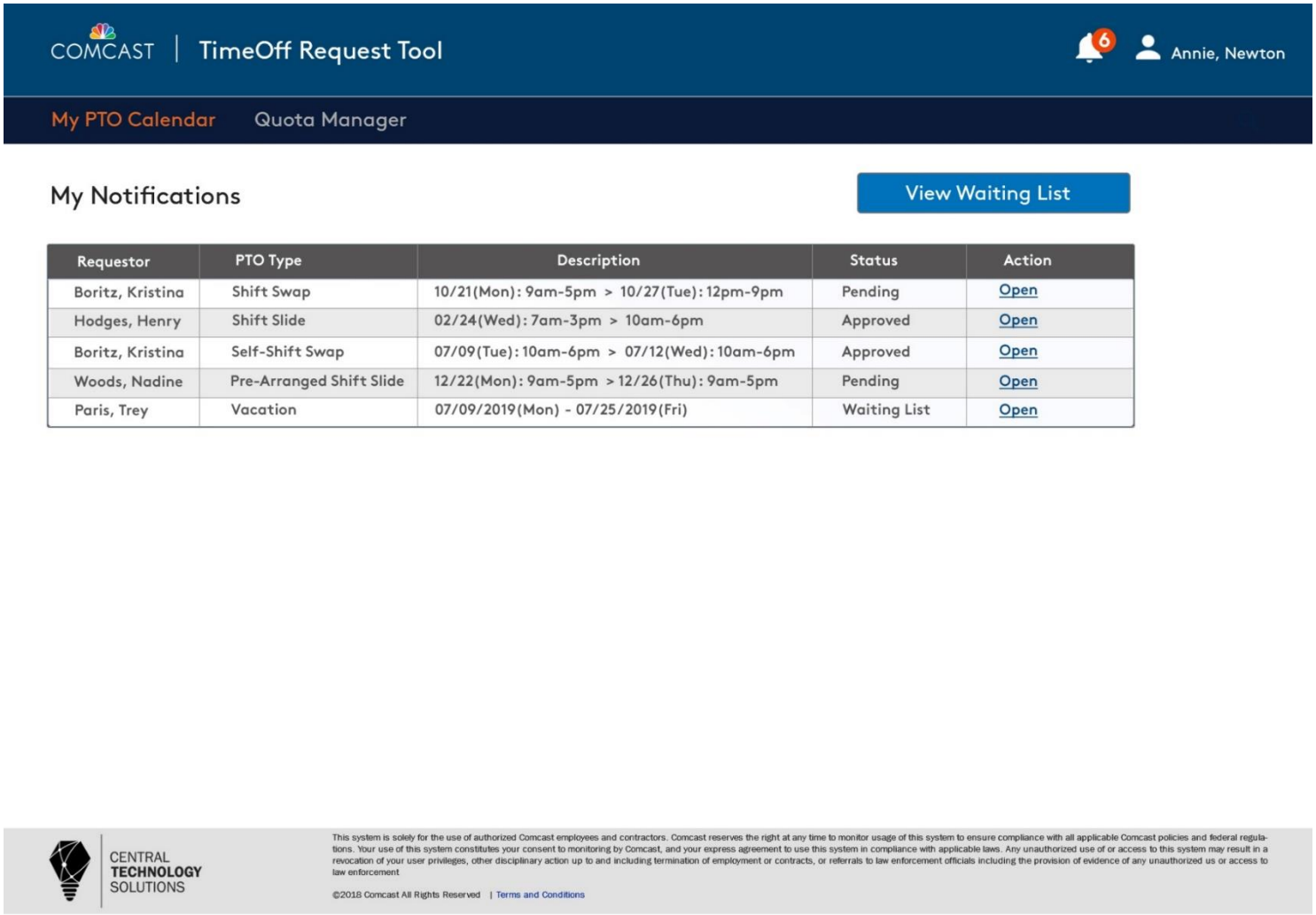
Shift Swap (with another Employee) - Schedule Flex (Employee View) :



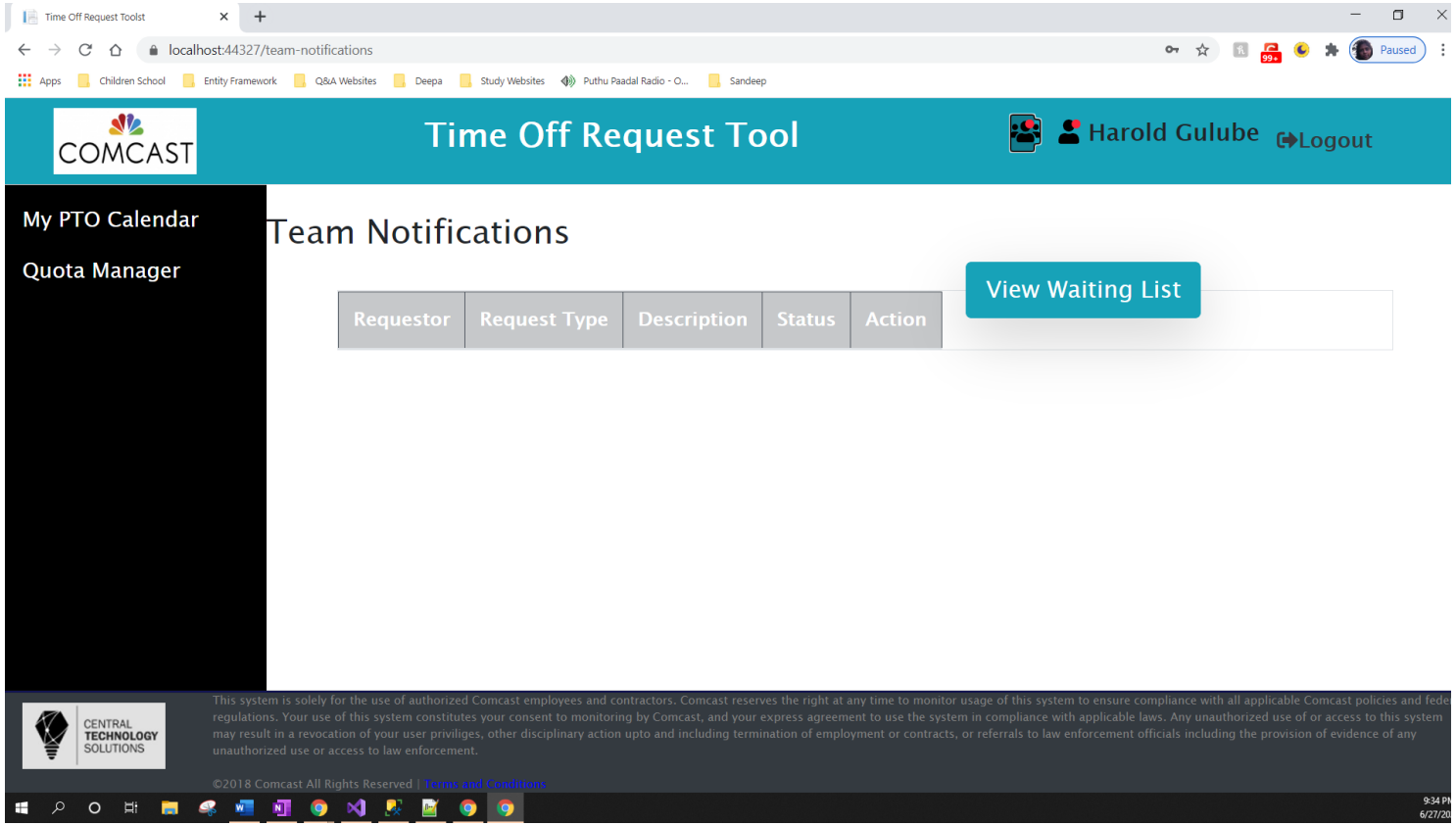
Application Image (Shift Swap) :


Time Off Request Toolst
localhost:44327/pto-calendar
Paused
Apps
Children School
Entity Framework
Q&A Websites
Deepa
Study Websites
Puthu Paadal Radio - O...
Sandeep
COMCAST
Time Off Request Tool
Harold Gulube
Logout
My PTO Cal
Quota Mana
gle weekends
week
day
Sat
6
13
20
21
22
23
24
25
26
27
New Schedule Flexibility Options
Select your Flex option
Shift Slide
Pre-Arranged Shift Slide
Self-Shift Swap
Shift Swap Request
Shift Swap
Trade your shift with a co-worker
Co-worker's Name
Please select your Co-worker.
Choose your co-worker's shift
Shift Start Date
mm/dd/yyyy
Start Time
--:-- --
End Time
--:-- --
Choose your own shift
Shift Start Day
2020-06-27
Start Time
12:00 AM
End Time
12:00 AM
Request Shift Swap
Cancel
9:26 PM
6/27/2020

My Notifications – Manager View :

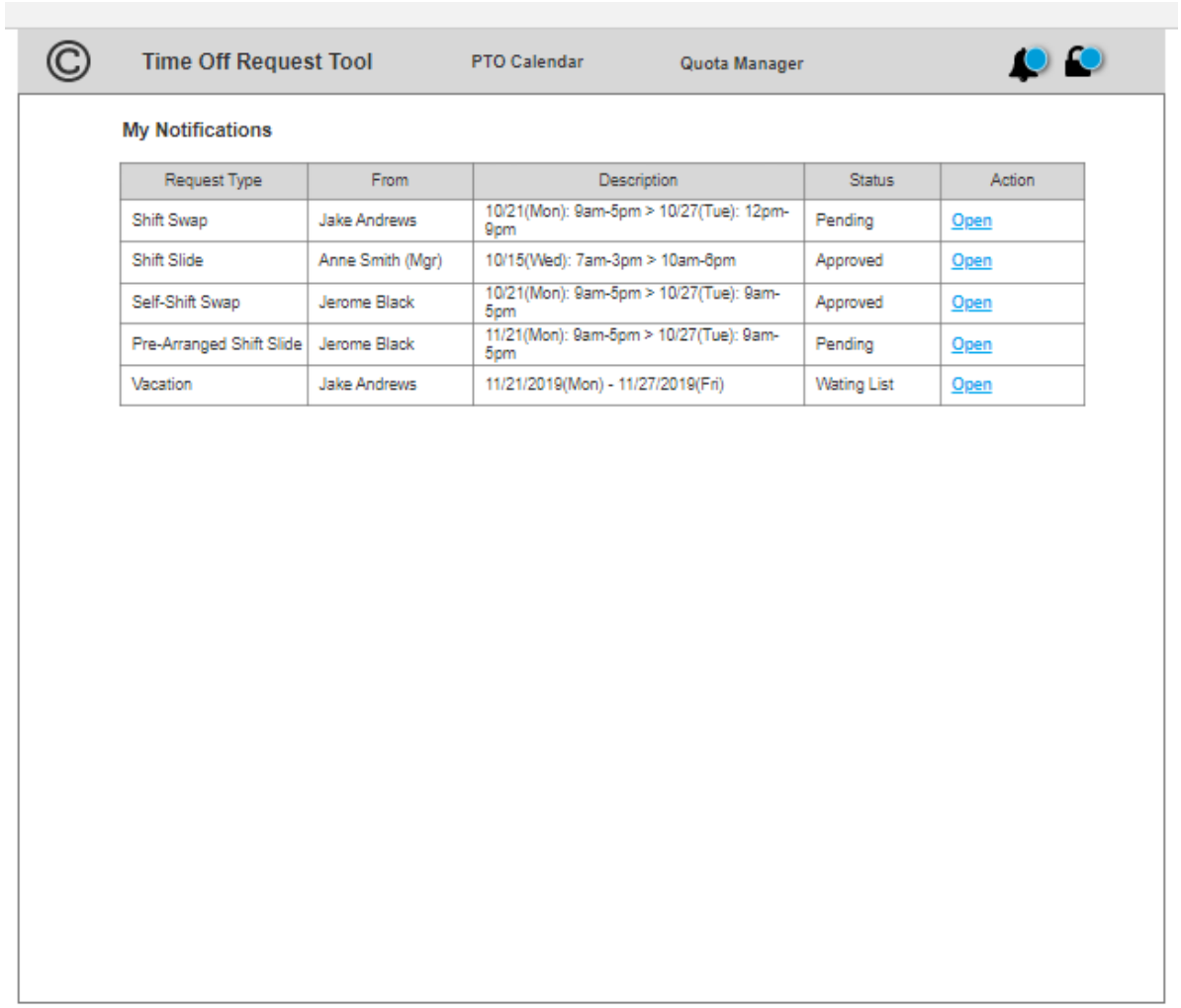


Application Image (Team Notifications) :

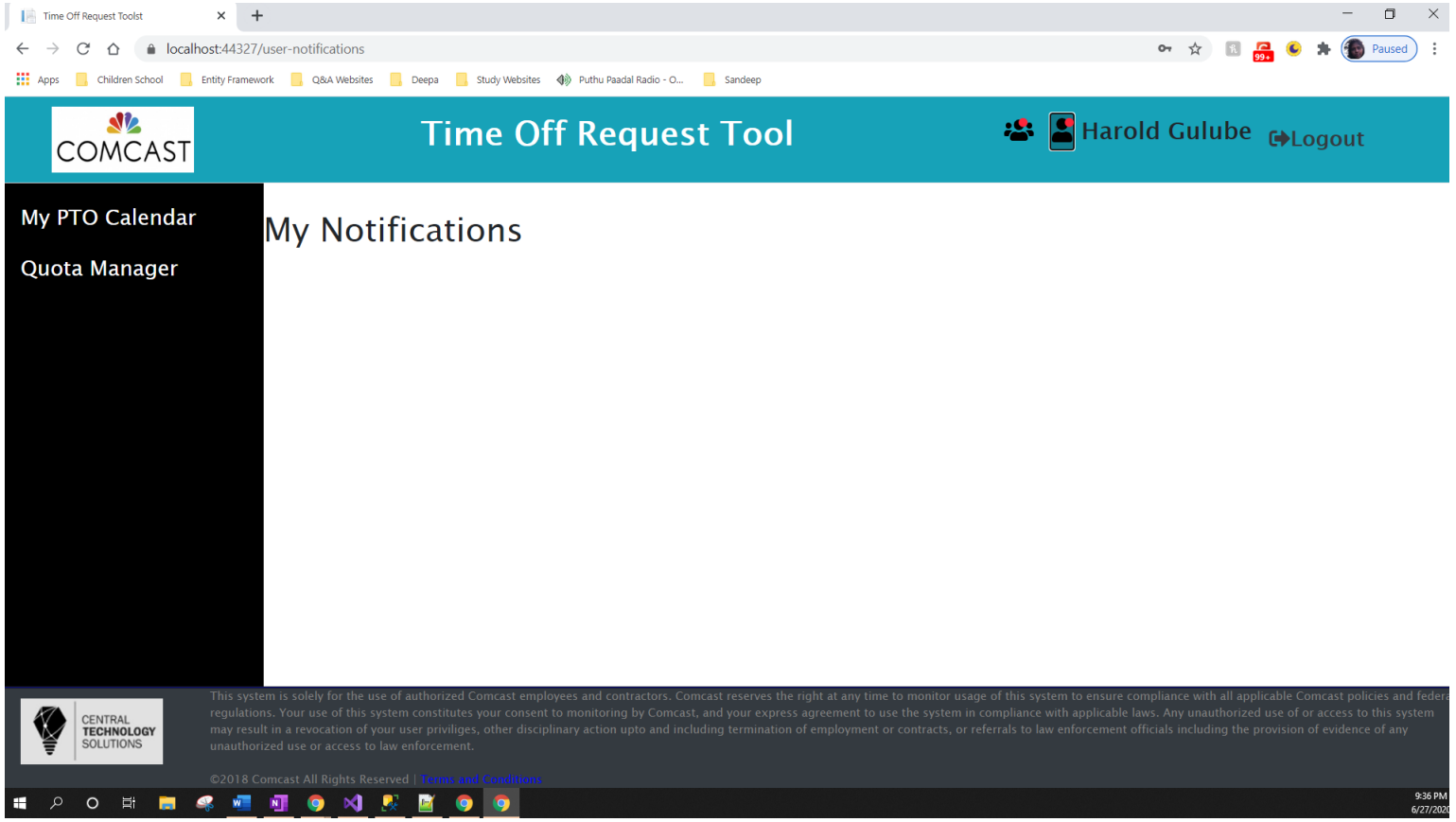


My Notifications (Employee View) :

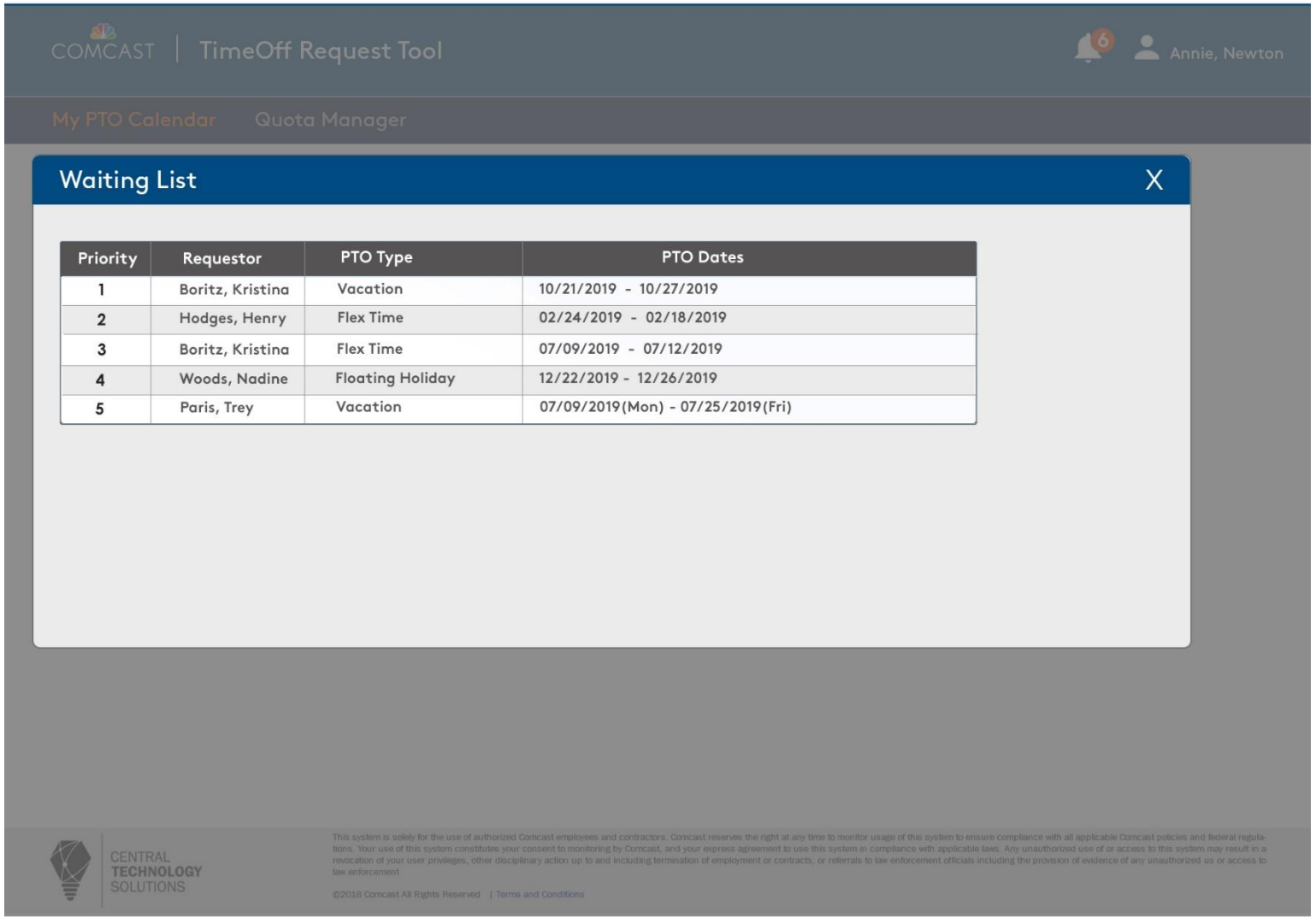
(Same as Manager View except “View Waiting List” button is not in Employee View)



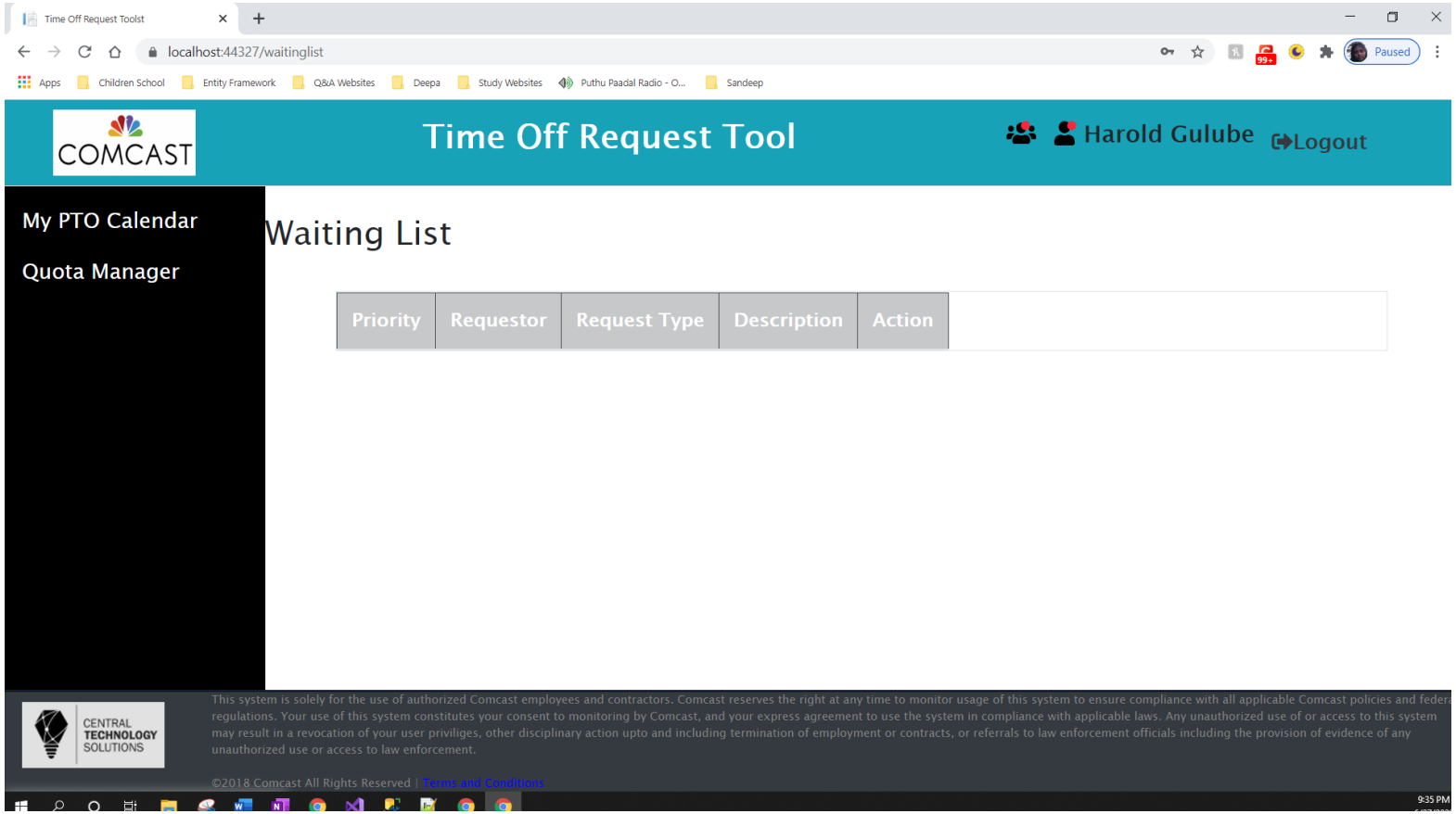
Application Image (My Notifications) :



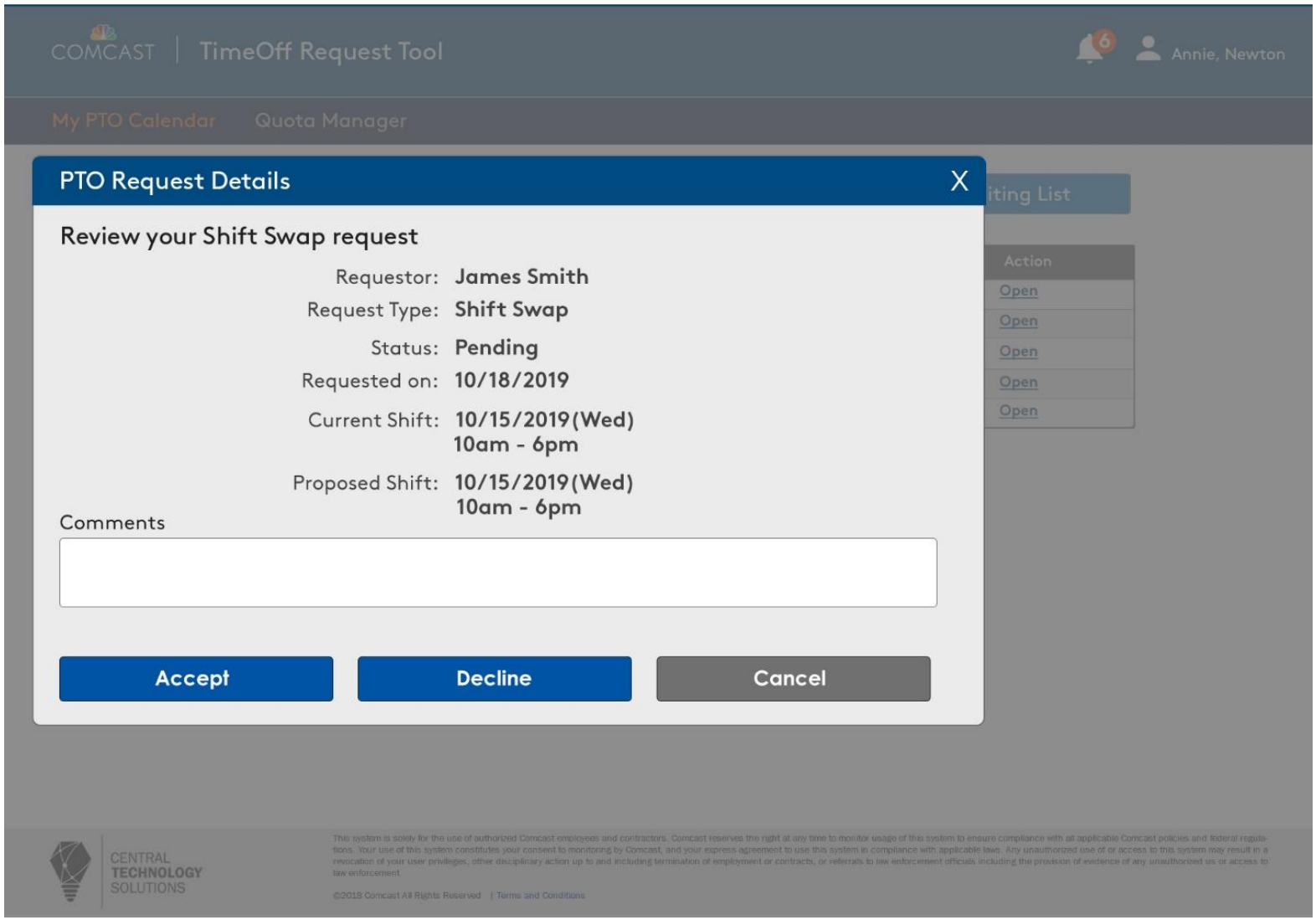
Waiting List (Manager View) :


COMCAST | TimeOff Request Tool
6
Annie, Newton
My PTO Calendar
Quota Manager
Waiting List
X
Priority
Requestor
PTO Type
PTO Dates
1
Boritz, Kristina
Vacation
10/21/2019 - 10/27/2019
2
Hodges, Henry
Flex Time
02/24/2019 - 02/18/2019
3
Boritz, Kristina
Flex Time
07/09/2019 - 07/12/2019
4
Woods, Nadine
Floating Holiday
12/22/2019 - 12/26/2019
5
Paris, Trey
Vacation
07/09/2019(Mon) - 07/25/2019(Fri)
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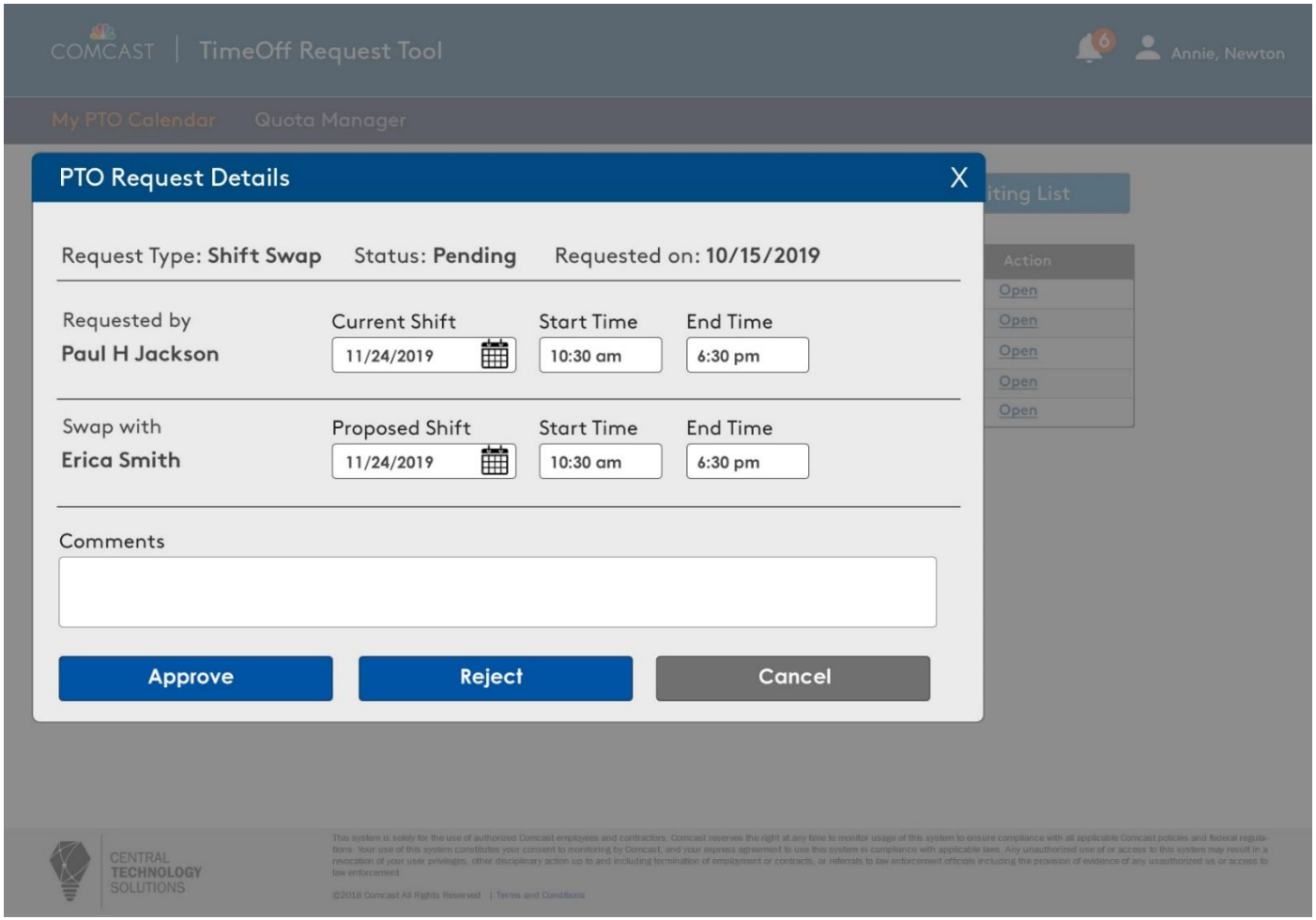
Application Image (Waiting List) :



My Notifications Details (Shift Swap Request) - Employee View :


COMCAST | TimeOff Request Tool
6
Annie, Newton
My PTO Calendar
Quota Manager
PTO Request Details
X
Review your Shift Swap request
Requestor: James Smith
Request Type: Shift Swap
Status: Pending
Requested on: 10/18/2019
Current Shift: 10/15/2019(Wed) 10am - 6pm
Proposed Shift: 10/15/2019(Wed) 10am - 6pm
Comments
Accept
Decline
Cancel
iting List
Action
Open
Open
Open
Open
Open
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My Notifications Details (Shift Swap Request) – Manager View :

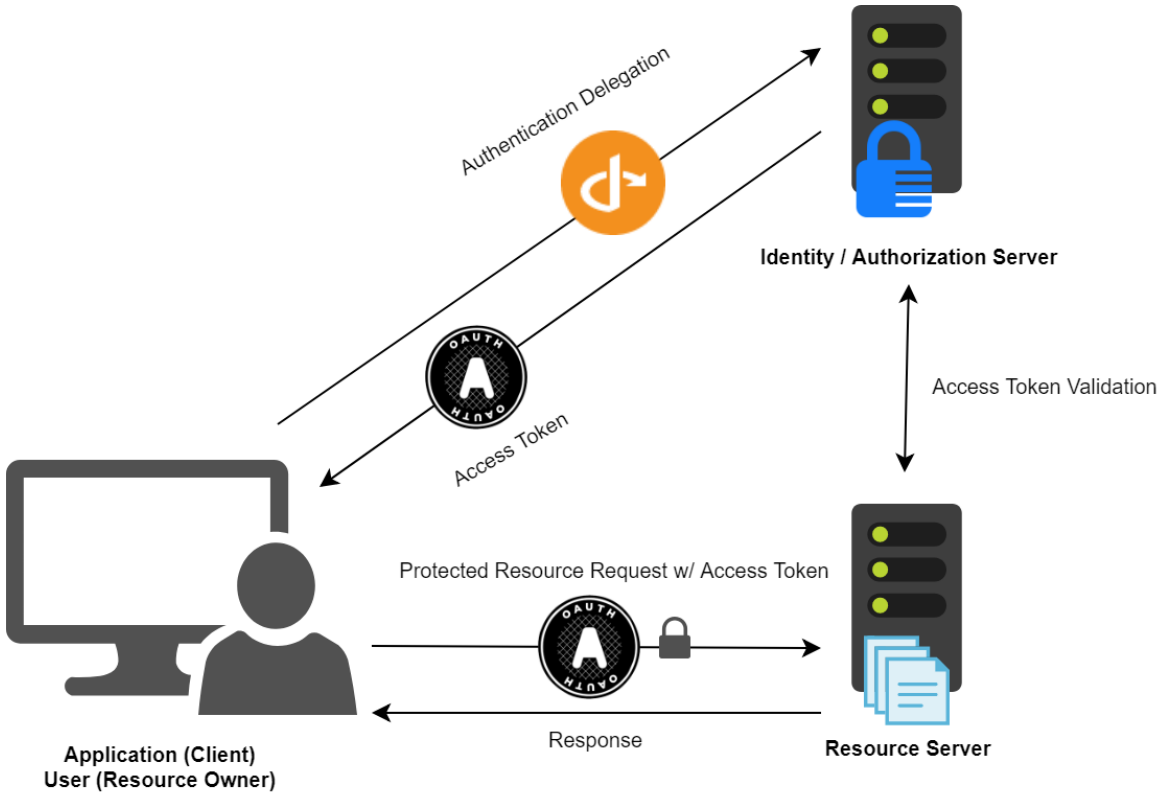


Project Share point Site :

https://appdevprojects.cable.comcast.com/sites/DefaultCollection/PTO%20Time%20Management/Shared%20Documents/Forms/AllItems.aspx

Diagram showing concepts and protocols fit together in a basic authentication and authorization scenario :

OIDC/OAuth flow



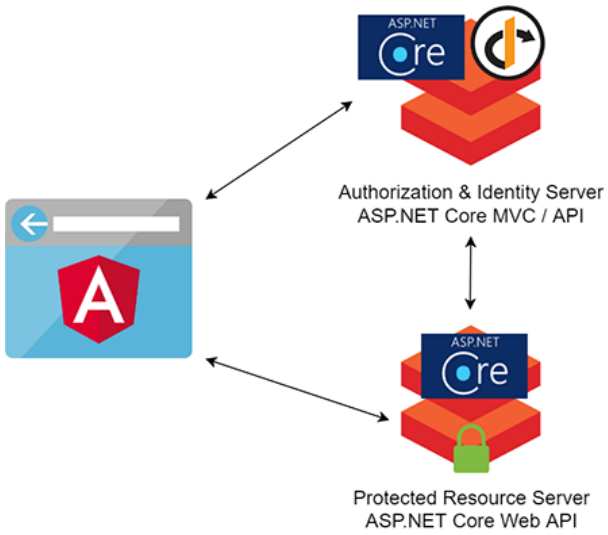
To summarize this workflow:

1. The user (resource owner) initiates an authentication request with the authorization server.
2. If the credentials are valid and everything checks out the authorization server obtains end-user consent and grants the client application an access token.
3. The access token is attached to subsequent requests made to the protected resource server.
4. The authorization server validates the access token; if successful the request for protected resources is granted, and a response sent back to the client application.

Architecture

Our solution architecture has three main components:

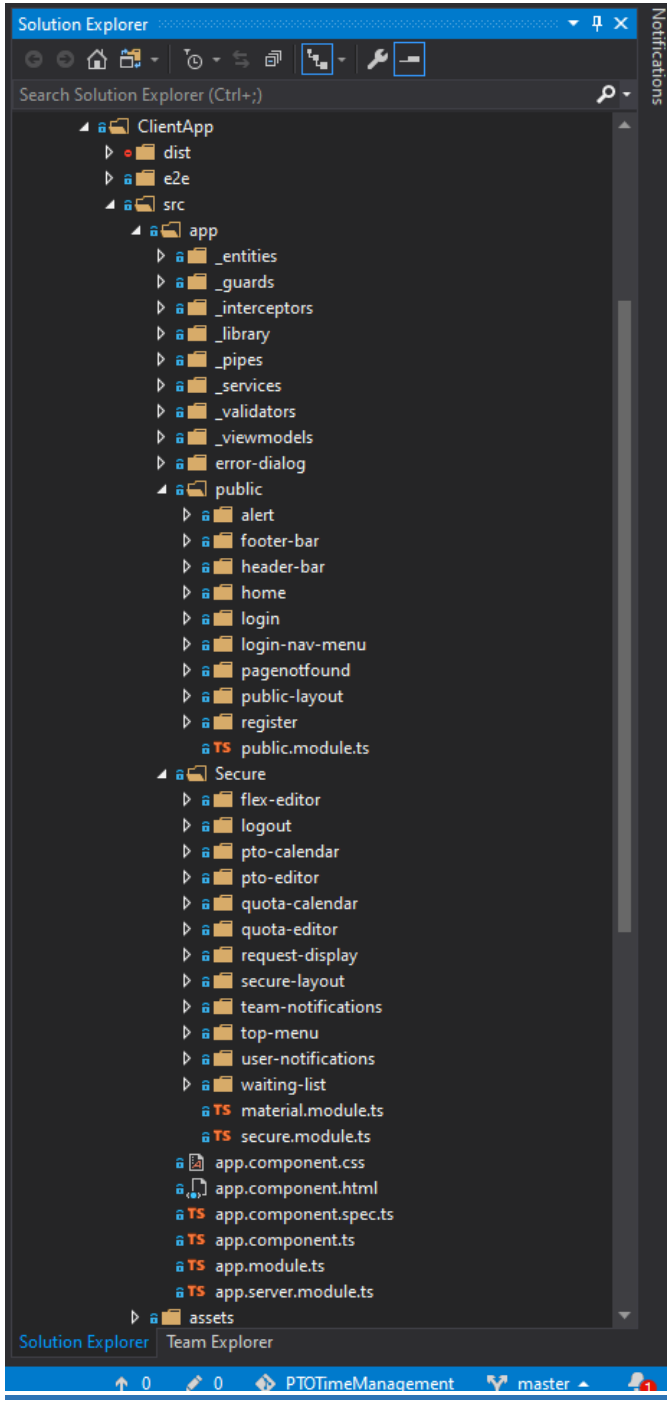
- SPA client application - Angular
- Authorization/Identity server - ASP.NET Core MVC and IdentityServer4
- Resource server - ASP.NET Core Web API

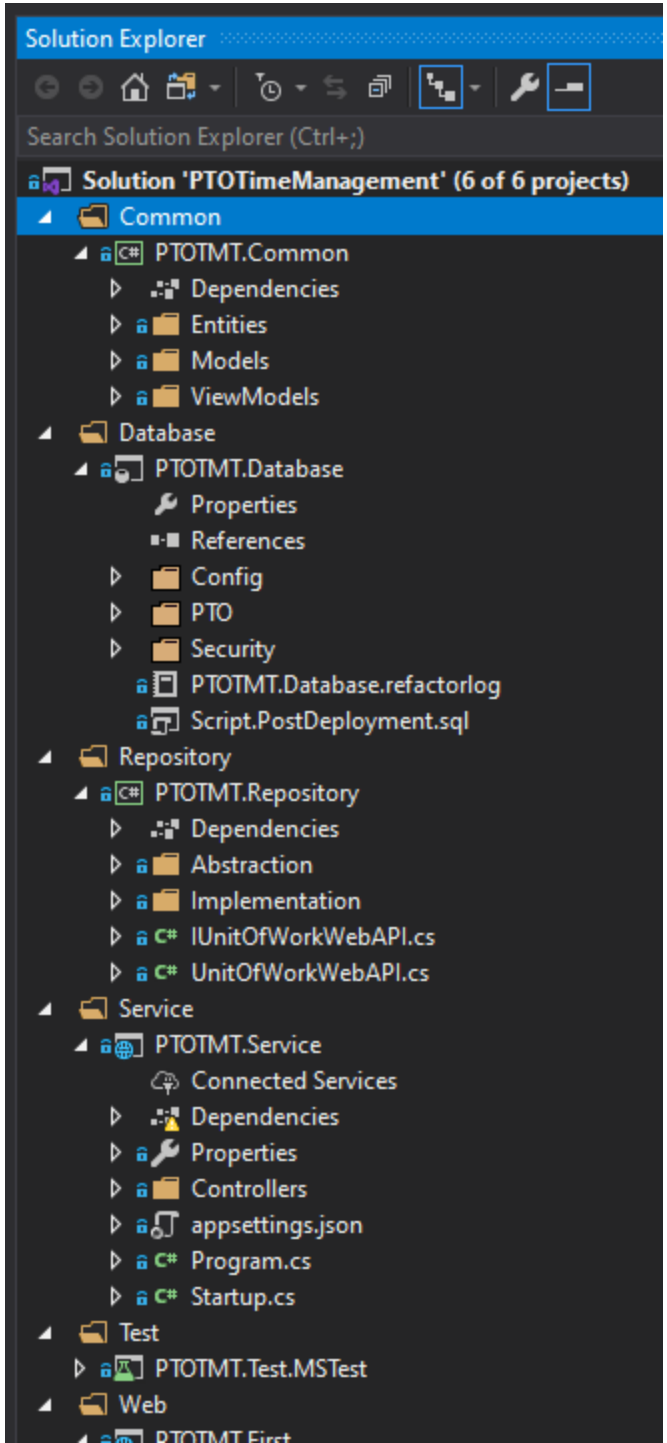


1. `OpenID Connect and OAuth 2.0 Framework for ASP.NET Core :`
   ```
   PM> Install-Package IdentityServer4
   ```

2. `ASP.NET Core Identity Integration for IdentityServer4 :`

```
PM> Install-Package IdentityServer4.AspNetIdentity
```


Solution Explorer
Search Solution Explorer (Ctrl+;)
ClientApp
dist
e2e
src
app
_entities
_guards
_interceptors
_library
_pipes
_services
_validators
_viewmodels
error-dialog
public
alert
footer-bar
header-bar
home
login
login-nav-menu
pagenotfound
public-layout
register
public.module.ts
Secure
flex-editor
logout
pto-calendar
pto-editor
quota-calendar
quota-editor
request-display
secure-layout
team-notifications
top-menu
user-notifications
waiting-list
material.module.ts
secure.module.ts
app.component.css
app.component.html
app.component.spec.ts
app.component.ts
app.module.ts
app.server.module.ts
assets
Solution Explorer
Team Explorer
0
0
PTOTimeManagement
master
Notifications


Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'PTOTimeManagement' (6 of 6 projects)
Common
PTOTMT.Common
Dependencies
Entities
Models
ViewModels
Database
PTOTMT.Database
Properties
References
Config
PTO
Security
PTOTMT.Database.refactorlog
Script.PostDeployment.sql
Repository
PTOTMT.Repository
Dependencies
Abstraction
Implementation
IUnitOfWorkWebAPI.cs
UnitOfWorkWebAPI.cs
Service
PTOTMT.Service
Connected Services
Dependencies
Properties
Controllers
appsettings.json
Program.cs
Startup.cs
Test
PTOTMT.Test.MSTest
Web